

Performance and Robustness Benchmarking of Battery SOC Estimation Methods for Embedded Applications

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Abstract

Robustness under disturbed sensing and signal-integrity faults is central to battery State-of-Charge (SOC) estimation. This study benchmarks four representative SOC-estimation approaches under one causal and cell-disjoint protocol: a direct-measurement model (DM) implemented as fixed-capacity Coulomb counting, a hybrid direct-measurement model (HDM) combining Coulomb counting with data-driven SOH adaptation, a hybrid equivalent-circuit model (HECM) based on a second-order RC observer, and a data-driven neural model (DD) implemented as a recurrent estimator. LFP aging trajectories cover multiple load, thermal, and SOH conditions. The evaluation combines three estimator-performance dimensions with an embedded implementation assessment based on numerical equivalence, inference latency, flash occupancy, and peak runtime RAM. Accuracy describes performance under nominal inputs. Robustness measures resistance to error growth when measurements are corrupted or unavailable. Recovery characterizes how quickly reliable SOC is restored after an initialization mismatch. Nominal cell-macro MAE is 0.0690 for DM, 0.0412 for HDM, 0.0337 for HECM, and 0.0258 for DD. The DD representative achieves the lowest nominal error and leads several disturbance cases, whereas HECM provides the strongest observed recovery profile. Individual scenarios therefore

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produce different model rankings. On the STM32H753, all four isolated SOC implementations reproduce their software references, remain within the available memory, and execute within the one-second sampling interval. Their latency and memory requirements differ substantially, demonstrating that deployment cost must be considered alongside estimator performance. The results show that nominal accuracy alone provides an incomplete basis for model selection and must be complemented by robustness, recovery, and embedded feasibility.

Keywords: Battery management system, State of charge, Robustness benchmark, Fault tolerance, Equivalent circuit model, Recurrent neural network, Embedded implementation, Microcontroller

1 Nomenclature

2 *Acronyms*

3	BMS	Battery management system
4	SOC	State of charge
5	SOH	State of health
6	DM	Direct-measurement model
7	HDM	Hybrid direct-measurement model
8	HECM	Hybrid equivalent-circuit model
9	DD	Data-driven neural model
10	ECM	Equivalent circuit model
11	EKF	Extended Kalman filter
12	MAE	Mean absolute error
13	RMSE	Root-mean-square error
14	ADC	Analog-to-digital converter
15	CV	Constant-voltage phase
16	OCV	Open-circuit voltage
17	GRU	Gated recurrent unit

18	LSTM	Long short-term memory
19	LFP	Lithium iron phosphate
20	<i>Symbols</i>	
21	t	Time
22	$I(t)$	Current signal at time t
23	$U(t)$	Voltage signal at time t
24	$T(t)$	Temperature signal at time t
25	$\widehat{\text{SOC}}(t)$	Estimated state of charge at time t
26	$\widehat{\text{SOH}}(t)$	Estimated state of health at time t
27	$\text{SOC}_{\text{ref}}(t)$	Dataset SOC ground truth at time t
28	$\text{SOH}_{\text{ref}}(t)$	Reference SOH target at time t
29	b_I	Multiplicative current-gain error
30	ΔU	Voltage bias or offset
31	ΔT	Temperature bias or offset
32	σ_I	Standard deviation of injected current noise
33	σ_U	Standard deviation of injected voltage noise
34	σ_T	Standard deviation of injected temperature noise
35	Q_c	Online cumulative charge state used by the SOC models
36	C_{nom}	Nominal capacity
37	C_{eff}	Effective capacity used inside the estimator
38	U_{max}	Upper voltage threshold used for full-charge detection
39	U_{tol}	Voltage tolerance used in the full-charge condition
40	U_1, U_2	Polarization voltages of the first and second RC branch
41	ΔMAE	Error increase relative to baseline

42 1. Introduction

43 1.1. Motivation and practical relevance

44 Reliable battery state estimation is a core BMS function because SOC and
45 SOH determine safety margins, usable energy, operating limits, charge con-
46 trol, and degradation-aware system management in both mobile and stationary
47 lithium-ion battery applications [1, 2, 3]. Practical deployment quality is not
48 defined by clean-condition accuracy alone, because real BMS operation is ex-
49 posed to sensor gain and offset errors, additive noise, initialization uncertainty
50 after restart, missing samples, irregular sampling, communication interruptions,
51 and transient signal spikes caused by measurement-chain limitations or distur-
52 bances in the surrounding system [4, 5, 6]. The relevant engineering question
53 is therefore not only whether an estimator achieves low MAE under nominal
54 measurements, but also whether it converges back to a reliable state after dis-
55 turbed initialization and whether it remains stable under physically plausible
56 measurement corruption.

57 As summarized conceptually in Figure 1, the present benchmark treats
58 deployment-facing estimator quality as a three-dimensional problem consisting
59 of nominal accuracy, recovery after state inconsistency, and robustness against
60 disturbed measurements and signal-integrity faults, embedded within broader
61 constraints on memory and execution time. The central question is how four
62 structurally different SOC-estimation classes, represented by the implementa-
63 tions studied here, respond to common failure mechanisms and microcontroller
64 constraints, and whether their nominal ranking remains informative under those
65 conditions. This distinction is essential because one implementation can be ac-
66 curate under clean measurements yet difficult to re-anchor, sensitive to a specific
67 input fault, or too costly under its trained inference semantics.

68 1.2. State of the art in battery state estimation

69 The literature commonly groups battery-state estimators into direct-measurement
70 or integration-based approaches, model-based approaches based on equivalent-
71 circuit or electrochemical models, hybrid approaches that combine physical

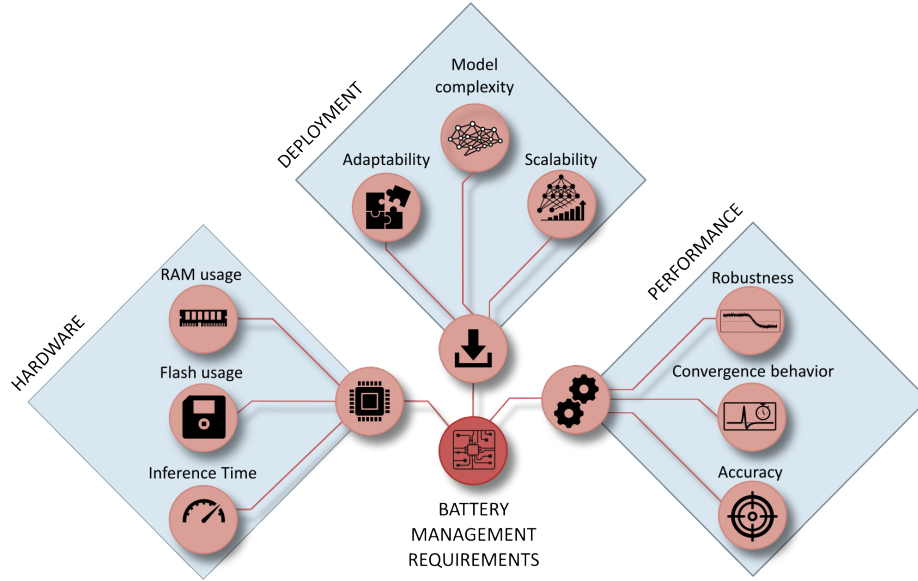


Figure 1: Conceptual overview of deployment-oriented requirements for embedded BMS state estimation. The broader requirement space includes implementation effort, hardware constraints, and estimator performance, whereas the present study focuses on the performance branch and evaluates baseline accuracy, convergence behavior, and robustness under measurement disturbances.

72 structure with learned components, and purely data-driven models based on
 73 neural sequence learning [7, 1, 2, 3]. Direct-measurement approaches, typically
 74 based on Coulomb counting, are attractive because they are simple, compu-
 75 tationally inexpensive, and easy to deploy. However, they remain structurally
 76 sensitive to current-gain error, missing data, and initialization errors because
 77 they lack an internal correction mechanism [7, 4]. Model-based approaches,
 78 especially ECM- and observer-based estimators, offer physical interpretability
 79 and voltage-based correction capability, but their practical behavior depends
 80 strongly on the quality of model parametrization, operating-point observability,
 81 and measurement quality [8, 2, 4].

82 Hybrid estimators seek to combine physical state propagation with learned
 83 adaptation of capacity, parameters, or auxiliary states, which has made joint
 84 SOC/SOH estimation an active research direction [9, 10, 11]. This combination

can improve baseline calibration, while it does not necessarily eliminate structural sensitivity to measurement disturbances. Data-driven approaches based on recurrent or sequence models can exploit nonlinear cross-relations between current, voltage, temperature, and derived online features, and recent work has also demonstrated their embedded feasibility through TinyML-oriented implementations, pruning, and quantization [12, 13, 14, 15, 3].

1.3. Performance and embedded deployment gap

A large share of the SOC estimation literature still emphasizes clean-data accuracy, average error metrics, or computational efficiency, while comparatively fewer studies analyze how estimators behave under realistic measurement faults and timing corruption [1, 13, 14, 15]. Existing robustness studies often remain confined to a single model family, for example observer-based robustness against modeling and measurement errors [4] or robust estimation concepts for specific BMS architectures under noise and uncertainty [5], whereas cross-family comparisons under the same disturbance protocol and under comparable embedded constraints remain limited. Benchmarking studies in the broader BMS context underline the importance of reproducible scenario design and deployment-relevant validation, but they do not by themselves close the gap between clean-data estimator comparison and measurement-level performance benchmarking across fundamentally different estimator classes [6]. This gap is especially relevant in embedded BMS design, where estimator selection must jointly consider nominal accuracy, convergence, robustness under disturbed measurements, and computational constraints.

1.4. Contribution and study objectives

This work addresses the gap through three linked questions: whether clean-data accuracy predicts performance under measurement and signal-integrity disturbances, which structural mechanisms govern degradation and recovery, and how the resulting behavior trades against embedded execution cost. To answer them, four deployment-oriented representatives are executed under one

Table 1: Estimator-class envelope and implementation scope. The last column defines the representative evaluated for each dominant SOC update and correction mechanism.

Class	Minimum defining mechanism	Broader class can include	Representative used in this study
Direct measurement	Causal integration of measured current using an assumed capacity	Coulombic efficiency, capacity adaptation, OCV or end-point anchoring, plausibility logic, and reset thresholds	Fixed-capacity Coulomb counting with a sustained full-charge CV reset and no voltage-residual correction
Hybrid direct measurement	Integration core augmented by an estimated or learned state/parameter	Learned capacity or efficiency, residual correction, multi-sensor fusion, and adaptive anchoring	Same Coulomb-counting and reset logic as DM, with effective capacity supplied by the shared causal LSTM-SOH estimate
ECM observer	Coulomb-counted SOC propagation plus a voltage model and feedback correction	Higher RC order, hysteresis, thermal states, SOC/SOH/temperature maps, and different nonlinear observers	Second-order RC model with EKF correction, charge/discharge SOC-SOH lookup surfaces, and shared causal SOH input
Data driven	Learned mapping from measured signals or their history to SOC	Feed-forward, convolutional, recurrent, attention-based, physics-informed, raw-signal, or engineered-feature models	Structurally pruned rolling-window GRU using voltage, current, temperature, SOH, Q_c , derivatives, and Δt

causal protocol on cell-disjoint LFP aging data covering multiple load and SOH states. The campaign combines physically interpretable measurement, timing, availability, quantization, and initialization disturbances with a common recovery criterion, cell-level uncertainty analysis, a paired SOH-input ablation, and STM32 measurements of the isolated SOC cores.

The four evaluated implementations provide distinct update and correction mechanisms whose behavior can be traced across the same disturbances, recovery criterion, and hardware measurements. The resulting comparison links estimator selection to the fault profile and resource constraints of the target BMS application.

2. Methodology

2.1. Estimator representatives and class boundaries

Figure 2 summarizes the causal benchmark chain, from measured signals and online feature reconstruction to disturbance injection and global and local evaluation. The four estimators were selected because their dominant SOC update and correction mechanisms differ. Table 1 relates these mechanisms to the broader design space and specifies the implementation evaluated for each class. Learned corrections, higher-order dynamics, richer sensing, and more complex sequence architectures extend these class envelopes while also changing deployment cost.

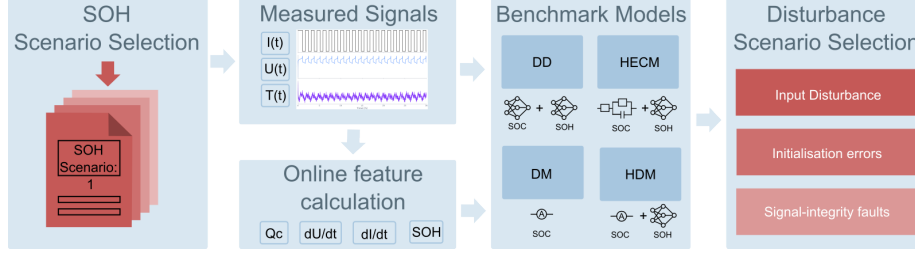


Figure 2: Methodology overview for the causal benchmark chain. Measured battery signals are converted into online auxiliary features, processed by representative estimator implementations, exposed to measurement and initialization interventions, and finally evaluated through global and local performance metrics.

134 All SOH-dependent representatives, namely HDM, HECM, and DD, receive
 135 the same causal LSTM-SOH trace so that differences arise from how the SOC
 136 branch consumes that context. DM is the low-complexity integration reference.
 137 HDM isolates the effect of learned capacity adaptation without adding volt-
 138 age feedback. HECM adds model-based voltage-residual correction to current-
 139 driven state propagation. DD represents compact temporal learning. The GRU
 140 was selected because SOC and disturbance recovery depend on recent signal his-
 141 tory while its gated single-state structure is lighter than a comparable LSTM.
 142 Because DD also consumes the engineered Q_c feature, it is a data-driven se-
 143 quence estimator with physics-informed inputs rather than a raw-signal-only
 144 neural model.

145 2.2. SOC estimation methods

146 2.2.1. Direct-measurement model (DM)

147 The direct-measurement SOC formulation used in DM, HDM, and as an
 148 explicit auxiliary feature in DD follows the same online cumulative-charge state,
 149 denoted here uniformly as Q_c . It is updated causally from the measured current
 150 stream as

$$Q_c(k) = Q_c(k-1) + \frac{I_k \Delta t_k}{3600 \text{ s h}^{-1}}, \quad (1)$$

151 where I_k is given in ampere and Δt_k in seconds, so that the factor 3600 s h^{-1}
 152 converts the increment to ampere-hours. The sign convention is fixed by the

153 benchmark implementation and the state is reset to $Q_c = 0$ after a sustained
 154 constant-voltage full-charge event. In the DM, this same charge state is mapped
 155 directly to SOC through the fixed nominal capacity

$$\widehat{\text{SOC}}_{\text{DM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{nom}}}, 0, 1\right). \quad (2)$$

156 2.2.2. Hybrid direct-measurement model (HDM)

157 The HDM preserves exactly the same Coulomb-counting core and differs
 158 only in the capacity term. The SOH estimate rescales the usable capacity as

$$C_{\text{eff}}(k) = C_{\text{nom}} \cdot \widehat{\text{SOH}}(k), \quad (3)$$

159 which yields

$$\widehat{\text{SOC}}_{\text{HDM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{eff}}(k)}, 0, 1\right). \quad (4)$$

160 This formulation makes the relation between DM and HDM explicit: both use
 161 the same online charge state Q_c , while only the denominator changes through
 162 online SOH support. In both direct-measurement variants, a sustained near-
 163 maximum voltage condition is interpreted as a constant-voltage full-charge event.
 164 If the voltage remains above $U_{\text{max}} - U_{\text{tol}}$, where U_{tol} denotes the admissible volt-
 165 age margin below the upper threshold U_{max} , for the configured CV duration,
 166 the internal charge state is reset so that $Q_c = 0$ and therefore $\widehat{\text{SOC}} = 1$. This re-
 167 flects the practical assumption that a completed CV phase provides a physically
 168 anchored full-charge reference.

169 2.2.3. Hybrid equivalent-circuit model (HECM)

170 The hybrid ECM model is implemented as a second-order RC model with
 171 the state vector

$$\mathbf{x}_k = \begin{bmatrix} \widehat{\text{SOC}}_k & U_{1,k} & U_{2,k} \end{bmatrix}^\top, \quad (5)$$

172 where U_1 and U_2 denote the polarization voltages of the two RC branches. The
 173 prediction step propagates the state according to

$$\mathbf{x}_k^- = \mathbf{A}_k \mathbf{x}_{k-1} + \mathbf{b}_k I_{k-1}, \quad (6)$$

174 with

$$\mathbf{A}_k = \text{diag}\left(1, e^{-\Delta t_k/\tau_1}, e^{-\Delta t_k/\tau_2}\right), \quad \mathbf{b}_k = \begin{bmatrix} \eta\Delta t_k/C_b \\ R_1(1 - e^{-\Delta t_k/\tau_1}) \\ R_2(1 - e^{-\Delta t_k/\tau_2}) \end{bmatrix}, \quad (7)$$

175 where $C_b = C_0\widehat{\text{SOH}}$ is the SOH-scaled available capacity. The corrected terminal-
176 voltage estimate is written as

$$\widehat{U}_k = \text{OCV}(\widehat{\text{SOC}}_k, \widehat{\text{SOH}}_k, I_k) + U_{1,k} + U_{2,k} + R_i I_k, \quad (8)$$

177 and the Kalman update uses the residual between $\widehat{U}_k$ and the measured terminal
178 voltage to correct the internal state. In the present implementation, the state
179 propagation is driven by the previous current sample I_{k-1} , whereas the mea-
180 surement equation uses the current sample I_k available at the correction step.
181 The ECM parameters are not constant. The implementation interpolates R_i ,
182 R_1 , R_2 , τ_1 , τ_2 , OCV, and dOCV/dSOC from a preidentified lookup table over
183 SOC and SOH, with separate charge and discharge parameter surfaces. The
184 HECM is implemented as a two-RC observer with SOH-dependent parameter
185 scheduling.

186 2.2.4. Data-driven SOC model (DD)

187 The data-driven SOC model is a GRU-based recurrent estimator with the
188 feature vector $\{U \text{ [V]}, I \text{ [A]}, T \text{ [}^\circ\text{C]}, \text{SOH}, Q_c, dU/dt, dI/dt, \Delta t\}$. The bench-
189 marked, structurally pruned and fine-tuned model consists of a single-layer GRU
190 with hidden size 67 followed by an MLP head with one hidden layer of size 96
191 and a sigmoid output layer. During optimization, the recurrent core is exposed
192 to causal sequence segments of length $L_{\text{SOC}} = 2024$ samples. Within this fea-
193 ture set, Q_c is exactly the same online cumulative-charge state defined in (1).
194 It is included explicitly so that the neural estimator receives the same charge-
195 throughput information that also drives DM and HDM. The derivative channels
196 are computed online as the finite differences $dI/dt(k) = (I_k - I_{k-1})/\Delta t_k$ and
197 $dU/dt(k) = (U_k - U_{k-1})/\Delta t_k$. They provide local transient information that is

not captured by the absolute channels alone. The additional feature Δt_k allows the network to distinguish amplitude changes from sampling-interval changes under irregular timing. During the primary benchmark, each output is calculated from the latest causal 2024-sample window and recurrent state is reset between windows, matching training and validation. This configuration uses measured and causally reconstructed inputs up to the current sample. The hardware analysis additionally evaluates continuous hidden-state forwarding as a deployment variant and reports its accuracy and resource demand separately.

2.3. SOH estimation method

The shared SOH estimator is an LSTM-based sequence-to-sequence model that operates on hourly aggregated online features derived from $\{U [V], I [A], T [^\circ C], EFC, Q_c\}$, where the aggregation uses the statistics mean, standard deviation, minimum, and maximum for each feature [16, 17, 18, 19]. This transforms the five base channels into a 20-dimensional hourly feature vector. Architecturally, the SOH network first projects the aggregated input through a two-layer feature embedding block with size 128, layer normalization, GELU activations, and dropout, then processes the sequence with a two-layer unidirectional LSTM of hidden size 112, and finally maps the recurrent output through three residual MLP blocks and a prediction head with hidden size 160 to obtain the SOH estimate for each hourly step.

During optimization, the SOH estimator is exposed to causal hourly sequence segments, and benchmark execution processes one hourly aggregate at a time with causal hidden- and cell-state forwarding. This allows the estimator to accumulate degradation context without access to future information. The first hourly output is anchored with the configured initial SOH value before subsequent hourly predictions are taken directly from the recurrent model output. The SOH prediction is updated at hourly cadence and held piecewise constant between update points. This online update structure matches the gradually evolving capacity fade in the dataset and supplies the latest SOH context to the faster SOC estimators. In the HDM, the predicted SOH enters the SOC

228 computation through the effective capacity relation $C_{\text{eff}} = C_{\text{nom}} \cdot \widehat{\text{SOH}}$, so the
 229 method remains structurally close to direct measurement while replacing the
 230 fixed-capacity assumption. In the HECM, the predicted SOH affects the capac-
 231 ity scaling and the lookup-based ECM parametrization, while in the DD model
 232 it is provided as an explicit SOC input feature that contextualizes the current
 233 measurement trajectory.

234 *2.4. Model and target-hardware preparation*

235 The software benchmark uses trained floating-point model parameters and
 236 input scalars fixed before holdout evaluation. Neural training, validation, hyper-
 237 parameter selection, scalar fitting, pruning, model selection, and HECM param-
 238 eter identification use the declared training and validation cells. The HPPC-
 239 derived HECM lookup surfaces were identified from this model-development
 240 subset and fixed without retuning before benchmark evaluation. The cells C09,
 241 C13, C15, C25, C27, and C29 are reserved for holdout evaluation. The compact
 242 SOC-GRU has hidden size 67 and the shared SOH-LSTM has hidden size 112.
 243 The continuous C29 lifecycle replay is identified separately as a mechanism-
 244 focused diagnostic.

245 The target-hardware experiment evaluates the isolated SOC cores on a NUCLEO-
 246 H753ZI board with a 480 MHz STM32H753 Cortex-M7. To separate SOC imple-
 247 mentation cost from the shared SOH service, the board receives the precomputed
 248 causal SOH trace used by the software reference. The SOH-LSTM itself is not
 249 executed on the microcontroller. Numerical equivalence and inference latency
 250 are measured directly on the board. Execution time is recorded for every valid
 251 inference with the processor-integrated hardware cycle counter, and each cell
 252 sequence is replayed three times. Flash occupancy is calculated from the com-
 253 piled firmware regions that must be stored in the microcontroller’s non-volatile
 254 program memory.

255 Runtime RAM is divided into statically allocated variable storage, dynam-
 256 ically allocated memory, and temporary memory used during nested function
 257 calls. The static component includes both initialized variables and variables

258 set to zero during system startup. Maximum temporary function-call mem-
 259 ory is measured by filling the available stack region with a known pattern be-
 260 fore initialization. After a representative 2025-sample replay of cell C09, the
 261 deepest overwritten address identifies the maximum function-call memory used.
 262 Dynamic-memory management is instrumented to record the maximum cumu-
 263 lative allocation reached during execution. The reported peak RAM combines
 264 these three contributions and covers startup, protocol handling, and estimator
 265 execution. An unused compilation-time memory reserve is kept separate from
 266 the measured demand. The exact rolling-window DD produces two valid calls
 267 after its 2023-sample warm-up in this memory replay, so its measured stack
 268 maximum applies to those calls, while its dominant static window allocation
 269 is determined at link time. Predictions are compared with both the software
 270 implementation and dataset SOC. These measurements quantify the resource
 271 demand of the isolated SOC implementations with SOH supplied as an external
 272 causal trace.

273 *2.5. Robustness benchmark design*

274 The primary cross-model benchmark manipulates current, voltage, tempera-
 275 ture, sample availability, and sampling time, all of which are measured or recon-
 276 structed online by a BMS. A separate initialization branch acts on deployment-
 277 accessible estimator states. Parameter mismatch is excluded from this common
 278 ranking and examined separately through the HECM lookup-table sensitivity.
 279 Figure 3 groups the primary scenarios into input disturbances, initialization
 280 errors, and signal-integrity faults, while Table 2 defines their implementation
 281 [6].

282 The disturbance magnitudes follow three practical levels of justification:
 283 deployment-level sensing limits, adverse measurement corruption, and signal-
 284 integrity stress conditions [20, 21, 22, 23, 24, 25]. Input disturbances include
 285 gain error, sensor noise, voltage spikes, and ADC quantization. The remaining
 286 scenarios address estimator initialization, sample availability, and sampling-time
 287 integrity. Together they form a controlled and reproducible benchmark matrix

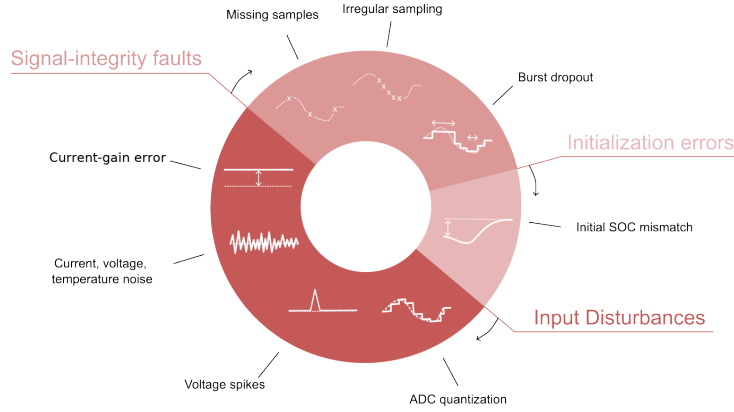


Figure 3: Taxonomy of the disturbance scenarios used in the performance benchmark, grouped into input disturbances, initialization errors, and signal-integrity faults.

288 for exposing estimator-specific failure mechanisms under a common protocol.

Table 2: Disturbance configurations used in the benchmark and their practical rationale. Exact numeric values are fixed benchmark settings. The cited sources motivate disturbance type and magnitude range.

Scenario	Benchmark setup	Practical origin / rationale
ADC quantization	$\Delta I = 0.01$ A, $\Delta U = 0.005$ V, and $\Delta T = 0.5$ °C	ADC steps, sensor scaling, rounding, and coarse deployment-level discretization [20, 21, 26]
Current gain error	$I_{\text{meas}} = I(1 + b)$ with $b = \pm 0.5\%$, $\pm 1.5\%$, and $\pm 3.0\%$. Both signs are evaluated at each magnitude	Sensor gain error, miscalibration, drift, and upper bound as stress case [21, 22, 23]
Current noise	Gaussian with $\sigma_I = 0.02$ A and 0.10 A	Sensing-chain noise, EMI pickup, and low and high sensor-noise levels [4, 5]
Voltage noise	Gaussian with $\sigma_U = 0.01$ V	Wiring noise, front-end noise, poor filtering, and EMI [4, 20, 24]

Table 2: Disturbance configurations used in the benchmark and practical rationale (continued).

Scenario	Benchmark setup	Practical origin / rationale
Temperature noise	Gaussian with $\sigma_T = 1.0^\circ\text{C}$	Sensor tolerance, conversion noise, thermal gradients, and sensor-to-cell mismatch [26, 11]
Voltage offset	Additive $U_{\text{meas}} = U + 0.02\text{ V}$	Front-end offset, calibration residual, wiring contribution, and adverse sensing stress
Temperature offset	Additive $T_{\text{meas}} = T + 3^\circ\text{C}$	Sensor placement error, calibration residual, and cell-to-sensor thermal mismatch
Initial SOC error	Initial mismatch of -0.10 SOC, with DD perturbed through online Q_c	Restart, storage, missing recent history, and weak LFP OCV anchoring [4, 8]
Missing samples	Every 50th sample frozen and separate random 2% loss	Packet loss, blocked updates, logger dropouts, and periodic and random data freeze [5, 25]
Irregular sampling	Uniform jitter: $\pm 0.1\text{ s}$, $\pm 0.5\text{ s}$, and $\pm 0.9\text{ s}$ around nominal 1 s grid	Scheduling jitter, bus latency, and asynchronous acquisition [6, 5, 25]
Burst dropout	One frozen 3600 s interval selected by maximum absolute net charge, with at least 12 h before and 24 h after the interval	Communication interruption, frozen task, stuck data stream, and measurement-only severe case [6, 5]
Voltage spikes	Single-sample spikes of $\pm 0.20\text{ V}$ every 1000 samples (1000 s)	EMI, switching disturbance, and corrupted voltage samples [24]

2.6. Metrics and evaluation criteria

For deployment-oriented interpretation, the benchmark results are synthesized along three performance dimensions: nominal accuracy under baseline con-

292 ditions, recovery after initialization mismatch, and robustness under sustained
 293 or transient measurement corruption and signal unavailability. Disturbance-
 294 related global performance is quantified through MAE, RMSE, bias, maximum
 295 error, and the 95th percentile absolute error, because these metrics provide a
 296 compact view of average performance, systematic deviation, and tail behavior
 297 across the selected evaluation windows. Local disturbance performance is an-
 298 alyzed through scenario-specific metrics such as jump counts, output variance,
 299 absolute-error variance, and excess rolling MAE, because several disturbances do
 300 not primarily manifest as window-wide MAE collapse but as short-term output
 301 fluctuations, delayed drift, or slow re-synchronization. Whenever a disturbance
 302 mask exists, the analysis further separates pre-disturbance, disturbed, and post-
 303 disturbance behavior in order to quantify disturbance-window error growth and
 304 residual post-disturbance error.

305 ADC quantization, current-gain error, voltage and temperature disturbances,
 306 missing samples, and irregular sampling are interpreted primarily through global
 307 metrics. Burst dropout, voltage spikes, current-noise detail, and initialization
 308 mismatch additionally use dedicated local evidence because their most relevant
 309 effects are episodic. The canonical initialization-recovery analysis compares each
 310 perturbed estimator trajectory with its cell-, window-, model-, SOH-, and seed-
 311 matched correctly initialized trajectory. The resulting paired trajectory differ-
 312 ence is the recovery signal, while absolute error against dataset SOC remains
 313 the accuracy measure. A first qualifying entry occurs when the paired differ-
 314 ence enters a 0.02 SOC band for at least 300 s. Persistent recovery is the first
 315 subsequent point after which the difference remains within this band for the
 316 rest of the 24-h observation horizon. Runs without this endpoint are right-
 317 censored. First-entry time, later departure from the band, censored and relapse
 318 fractions, and excess-error area under the curve provide complementary recov-
 319 ery diagnostics. The protocol maps a -0.10 SOC-equivalent mismatch into the
 320 coulomb-counting SOC initialization for DM and HDM, the EKF SOC state
 321 for HECM, and the capacity-equivalent Q_c offset for DD. The DD interven-
 322 tion passes through the recurrent network, so its realized paired difference is

323 recorded at the first commonly scored sample. Common observation starts at
324 source sample 2023, approximately 0.562 h after the intervention. Endpoints
325 already satisfied at this boundary are left-censored and recorded as conserva-
326 tive upper bounds. The comparison therefore measures observable re-anchoring
327 after deployment-accessible, estimator-specific initialization interventions.

328 Each holdout cell is treated as one independent statistical unit. Protocol-
329 defined SOH windows and random seeds are first summarized within each cell,
330 after which the six cells receive equal macro weight. Hierarchical bootstrap
331 resampling with seeds nested inside cells and 10,000 repetitions provides 95%
332 confidence intervals. Scenario effects are evaluated as paired, cell-matched dif-
333 ferences from baseline, and estimator pairs are compared using exact two-sided
334 sign-flip tests, raw mean and median differences, paired standardized effect sizes,
335 and Holm-adjusted p -values. Holm correction is applied within each reported
336 scenario family. With six independent cells, effect magnitudes and confidence
337 intervals provide the primary evidence and p -values support their interpretation.
338 The full machine-readable tests accompany the result tables, while the baseline
339 pair tests are reproduced in the Appendix.

340 The rolling DD implementation first emits an SOC estimate at source sample
341 2023 because it requires a 2024-sample causal input window. All global accuracy,
342 robustness, stratified, confidence-interval, and paired-test calculations therefore
343 use the common source-sample mask $k \geq 2023$ for every estimator. Each 24-
344 h window contributes 84,377 matched evaluation points per estimator, while
345 each 48-h dropout window contributes 170,777. The common mask gives every
346 estimator the same source interval and number of scored samples.

347 The decision synthesis provides a relative summary of the four implemen-
348 tations. Accuracy combines baseline MAE, RMSE, and 95th-percentile error.
349 Robustness assigns equal weight to eight evaluated disturbance families: sensor
350 noise, current-gain error, sensor offsets, quantization, missing samples, timing
351 jitter, burst dropout, and voltage spikes. The current-gain family uses the ad-
352 verse direction from the matched signed sweep. Non-positive ΔMAE values con-
353 tribute zero disturbance penalty. Levels within a family are averaged before the

family scores are combined, giving every disturbance family equal influence. Recovery combines observed persistent recovery-or-censor time, excess-error area, and persistent censored fraction from the canonical paired initialization analysis. Relapse after a first qualifying entry remains a separate diagnostic because it is defined only for runs that enter the recovery band. Equal-scenario and high-severity weighting variants quantify the sensitivity of the summary to alternative deployment priorities.

3. Experimental Setup

3.1. Dataset and data acquisition

The benchmark uses the MGFarm LFP 18650 dataset because its controlled design-of-experiments aging study provides long 1-Hz full-cell trajectories, repeated capacity checkups, multiple operating conditions, and enough independent cells for a cell-disjoint comparison [27, 28, 29]. Model development and holdout testing use a fixed split. SOC and SOH training use C01, C03, C05, C11, C17, and C23. Validation uses C07, C19, and C21. The independent holdout benchmark uses C09, C13, C15, C25, C27, and C29. Figure A.26 shows the complete SOH trajectories together with their charge-rate, discharge-rate, and depth-of-discharge aging conditions. Table A.4 identifies every cell and its split and reports measured duration, SOH, temperature, and load coverage. The six holdout cells are summarized separately in Figure A.21. This presentation makes both the cell-disjoint development boundary and the diversity of the test cells explicit. LFP’s flat mid-SOC OCV and chemistry-specific aging behavior are integral to the tested problem.

The holdout cells are assigned to descriptive load groups using the measured 95th-percentile absolute C-rate: C25 and C27 form the low-load group, C09, C13, and C15 the middle-load group, and C29 the high-load case. C29 provides an exploratory high-load trajectory, whereas the low- and middle-load groups support multi-cell summaries. Within each trajectory, performance is additionally stratified by reference SOH (fresh at $\text{SOH} \geq 0.90$, mid-life at

383 $0.80 \leq \text{SOH} < 0.90$, and aged at $\text{SOH} < 0.80$), instantaneous absolute C-
 384 rate (< 0.5 , $0.5\text{--}1.5$, and ≥ 1.5), temperature ($\leq 30^\circ\text{C}$, $30\text{--}35^\circ\text{C}$, and $> 35^\circ\text{C}$),
 385 and SOC (< 0.20 , $0.20\text{--}0.80$, and > 0.80). Unobserved cell-state combinations
 386 are marked as not available.

387 3.2. Dataset ground truth and preprocessing

388 The reference preprocessing first computes the signed transferred charge as
 389 $Q_k = I_k \Delta t_k / 3600$ and the absolute transferred charge as $|Q_k|$, from which the
 390 capacity during dedicated capacity-test intervals is obtained as $C_{\text{ref}} = \sum_k |Q_k|$.
 391 The reference SOH target is then defined offline as $\text{SOH}_{\text{ref}} = C_{\text{ref}} / C_{\text{ref},0}$, where
 392 $C_{\text{ref},0}$ denotes the first measured reference capacity of the trajectory, and the
 393 resulting capacity and SOH columns are interpolated between the dedicated
 394 capacity-test anchors. The dataset SOC ground truth used in the benchmark
 395 is reconstructed offline as $\text{SOC}_{\text{ref}} = 1 + Q_c / C_{\text{ref},0}$, where the cumulative charge
 396 state Q_c is propagated from the signed charge and reset to 0 at the upper
 397 voltage anchor and to $-C_{\text{ref},0}$ at the lower voltage anchor during preprocessing.
 398 This reconstructed SOC serves as the common dataset ground truth for every
 399 reported SOC error. It is an offline evaluation quantity because its dataset-side
 400 processing and anchor logic are not fully available to an online estimator during
 401 deployment. All nominal accuracy values therefore refer to this consistently
 402 applied dataset ground truth. Paired ΔMAE compares each disturbed run with
 403 its matched baseline under the same ground truth.

404 3.3. Test trajectory and benchmark scenarios

405 All estimators are evaluated on every available holdout-cell/SOH combina-
 406 tion under the same disturbance protocol. One representative 24-h window is
 407 frozen for each available fresh, mid-life, and aged state, yielding 16 windows.
 408 C27 contributes only a fresh window because its measured SOH does not fall
 409 below 0.90. Windows start at a measured full-charge anchor ($\text{SOC} \geq 0.98$,
 410 $U \geq 3.58\text{ V}$) and are selected as measured-feature medoids without using es-
 411 timator outputs or errors. The one-hour dropout uses 48 h from the same

412 anchor to retain approximately 24 h of post-gap observation. The shared SOH
 413 LSTM receives 192 h of undisturbed causal context before every scored window,
 414 matching its trained sequence length. Figure A.23 summarizes this protocol.

415 The benchmark contains 19 scenario definitions: the nominal baseline, two
 416 current-noise levels, voltage and temperature noise, three signed current-gain
 417 magnitudes, voltage and temperature offsets, ADC quantization, initial SOC
 418 mismatch, periodic and random sample loss, three timing-jitter levels, a one-
 419 hour burst dropout, and randomized-sign voltage spikes. Each current-gain
 420 magnitude contains matched positive and negative sign sublevels. Adverse-
 421 direction summaries report the larger paired ΔMAE of the two signs at each
 422 magnitude, capturing the larger sensor-gain penalty when signed compensation
 423 with baseline error occurs. Gaussian sensor-noise cases use 10 deterministic
 424 seeds. Random sample loss, jitter, and spike cases use 5 seeds. Deterministic
 425 cases run once. Figure A.22 maps the scenario definitions to the manipulated
 426 measurement or state and marks the selected paired reference-SOH ablations.
 427 Input disturbances act directly on measured channels, initialization interven-
 428 tions act on each estimator’s deployment-accessible state, and signal-integrity
 429 scenarios alter sample availability or timing while preserving causal execution.

430 3.4. Implementation details

431 Every estimator is executed causally, and all auxiliary quantities required
 432 by the models are rebuilt directly from disturbed inputs. This applies to Q_c ,
 433 EFC, dU/dt , dI/dt , and hourly SOH aggregates, ensuring that estimator inputs
 434 originate from the causal measurement path. The SOH LSTM forwards hidden
 435 and cell states across hourly updates after its 192-h initialization context. The
 436 primary SOC-GRU evaluation preserves its trained sequence-to-one semantics:
 437 each prediction uses the latest 2024-sample rolling window with recurrent state
 438 reset between windows. The continuous-stateful stream forms a separate de-
 439 ployment configuration whose full-trajectory accuracy and resource demand are
 440 reported alongside the rolling implementation. In freeze and dropout scenar-
 441 ios, disturbed or frozen measurements propagate consistently into derived online

442 features. The dropout start is selected from measured current as the eligible one-
 443 hour interval with the largest absolute integrated net charge. Selection enforces
 444 at least 12 h of pre-gap observations and 24 h of post-gap observations and re-
 445 mains independent of dataset SOC and estimator output. Direct-measurement
 446 estimators use the same CV-triggered reset that defines the full-charge start,
 447 whereas the DD initial-SOC test uses an equivalent offset of online Q_c because
 448 the network exposes no explicit SOC state. The environment enforces common
 449 nominal-capacity metadata, disturbed streams, targets, masks, and metrics.
 450 Model-specific capacity handling remains part of the estimator definitions.

451 3.5. HECM lookup-table sensitivity

452 A separate HECM-only analysis quantifies whether its current-gain response
 453 depends on exact lookup-table calibration. It applies one systematic perturba-
 454 tion at a time to the fixed charge and discharge surfaces. The R_i , R_1 , and R_2
 455 surfaces are jointly scaled by -10% or $+10\%$, while the OCV surfaces are shifted
 456 by -10 mV or $+10\text{ mV}$. The OCV derivative and time-constant surfaces remain
 457 fixed. Each of these four perturbations and the nominal lookup are evaluated
 458 under baseline current and matched -3% and $+3\%$ current-gain errors over all
 459 16 frozen windows, yielding 240 HECM runs. The analysis uses the scenario-
 460 matched causal SOH traces and common source-sample mask. Windows are
 461 averaged within each cell before equal-weight six-cell aggregation and a 10,000-
 462 draw cell bootstrap. For each lookup condition, the adverse gain penalty is the
 463 larger cell-level ΔMAE from the two gain signs. The lookup-gain interaction
 464 subtracts the nominal-lookup adverse penalty from the perturbed-lookup ad-
 465 verse penalty. These one-at-a-time shifts are controlled local sensitivity probes
 466 rather than probabilistic uncertainty bounds. They remain outside the cross-
 467 model robustness score.

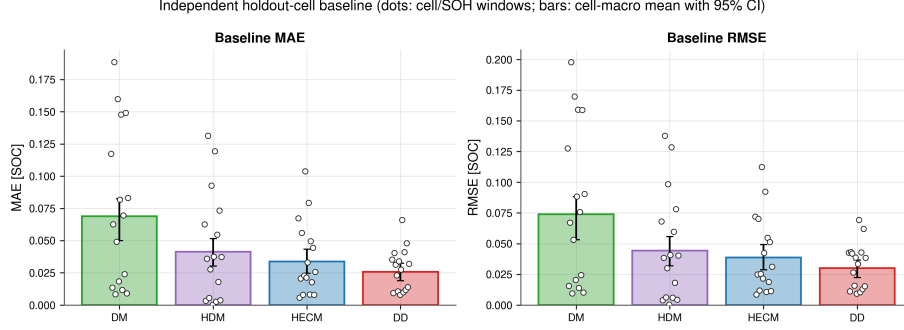


Figure 4: Baseline accuracy against the dataset SOC ground truth across six independent holdout cells. Dots show cell/SOH-window values. Bars and error bars show the equal-weight cell-macro mean and hierarchical 95% confidence interval. DD has the lowest macro MAE and RMSE, while DM has the largest error and between-cell spread.

4. Results

4.1. Baseline performance

Figure 4 summarizes nominal accuracy against the dataset ground truth across the 16 predefined windows on the common source-sample interval. Equal weighting of the six holdout cells gives the lowest baseline MAE to DD at 0.0258 (95% CI 0.0189–0.0325), followed by HECM at 0.0337 (0.0246–0.0434), HDM at 0.0412 (0.0301–0.0515), and DM at 0.0690 (0.0499–0.0823). The RMSE ranking is identical, with values of 0.0300, 0.0388, 0.0442, and 0.0740, respectively. Cell-level points reveal substantial between-cell variation, particularly for DM, and place the cell-macro means in the context of heterogeneous trajectories. Exact paired sign-flip tests support the magnitude and direction of the principal contrasts. With six independent cells, the Holm-adjusted values remain above 0.05, so confidence intervals and paired effect sizes provide the main statistical evidence reported in the Appendix. These values quantify accuracy against the consistently reconstructed dataset ground truth.

4.2. Current-gain sensitivity

Figure 5 summarizes the matched signed current-gain sweep at $\pm 0.5\%$, $\pm 1.5\%$, and $\pm 3.0\%$. For each magnitude and holdout cell, the adverse-direction sum-

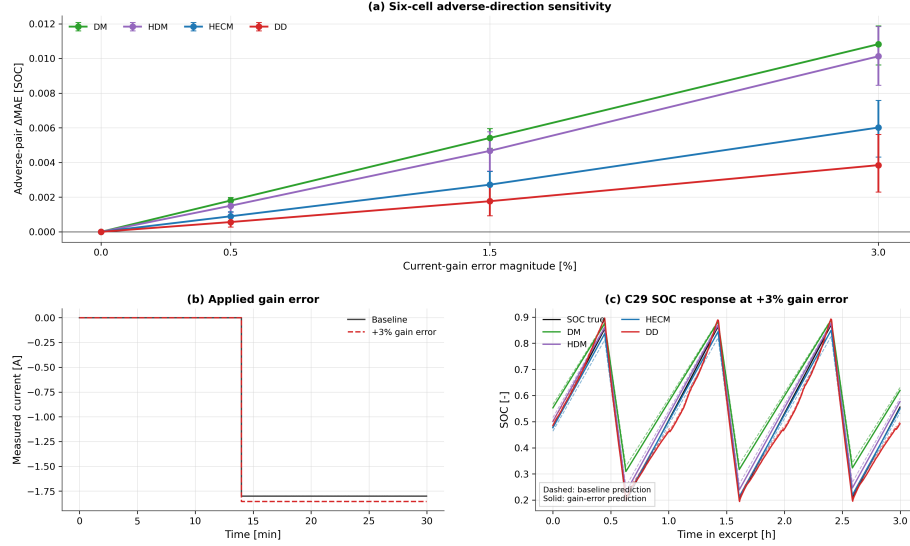


Figure 5: Current-gain sensitivity. (a) Six-cell adverse-direction ΔMAE for each matched $\pm b$ pair with hierarchical 95% confidence intervals. (b) Representative current before and after the +3% gain error. (c) C29 SOC trajectories over three charge–discharge intervals. The adverse-pair summary reports the larger penalty from each matched sign pair.

mary uses the larger MAE change from the two signs because one direction
can partially compensate a pre-existing baseline error. The resulting degra-
dation increases monotonically with gain-error magnitude for every estimator.
DM and HDM show the strongest response, HECM is less sensitive, and DD is
least sensitive. At 3%, the cell-macro MAE increases by 0.0108 for DM, 0.0101
for HDM, 0.0060 for HECM, and 0.0038 for DD. The complete signed sweep
remains available for interpreting direction-specific behavior.

The HECM lookup-table analysis in Figure A.25 separates baseline cali-
bration effects from current-gain robustness. The nominal lookup produces a
baseline MAE of 0.0337 and an adverse 3% current-gain penalty of 0.00602. Re-
sistance scaling changes baseline MAE to 0.0351 at -10% and 0.0327 at $+10\%$,
while the corresponding gain penalties remain 0.00614 and 0.00610. OCV shifts
change baseline MAE to 0.0314 at -10 mV and 0.0373 at $+10\text{ mV}$, while their
gain penalties are 0.00628 and 0.00580. Relative to the nominal lookup, the

four macro interaction effects range only from -0.00022 to $+0.00026$ MAE, and every 95% interval includes zero. Within these local perturbation ranges, lookup calibration affects absolute HECM accuracy more than its incremental current-gain penalty.

The continuous C29 analysis in Figure 6 complements the window-based benchmark and separates three mechanisms that independent 24-h windows cannot resolve. The gain-error contribution varies over life because its signed Coulomb-counting error interacts with charge direction and operating state. Between full charges, the penalty often grows within an interval, but full-charge anchoring can partially reduce or restructure it. The process is therefore neither an unrestricted monotonic drift nor a complete reset. Across this single full-life replay, the adverse 3% gain error increases MAE by 0.0102 for DM, 0.0072 for HDM, 0.0037 for HECM, and 0.0048 for DD. HECM therefore has the smallest full-life gain-error contribution in C29, while DM and HDM remain most exposed. This diagnostic is not a six-cell lifecycle ranking.

4.3. Noise sensitivity

Repeated sensor noise produces smaller global penalties than persistent current gain error, with a response that depends on the disturbed channel and estimator structure. At $\sigma_I = 0.10$ A, HECM has the largest mean macro penalty ($\Delta\text{MAE} = 0.0038$), although its interval spans zero. DM changes by -0.0009 , while HDM and DD remain near zero. These small signed changes arise from finite-window compensation around imperfect baseline trajectories. Voltage noise ($\sigma_U = 0.01$ V) produces a 0.0014 penalty for DM and 0.0013 for HDM, while HECM and DD remain below 0.0003. Temperature noise is neutral for DM and small for the SOH-aware models. Figure 7 shows the local transmission of current noise, while Figure 8 reports repeated six-cell effects with hierarchical uncertainty.

The offset cases expose a different pattern from zero-mean noise. A $+3^\circ\text{C}$ temperature offset increases DD MAE by 0.0074 (95% CI 0.0027–0.0125), while the corresponding changes for DM, HDM, and HECM are 0.0000, 0.0002, and

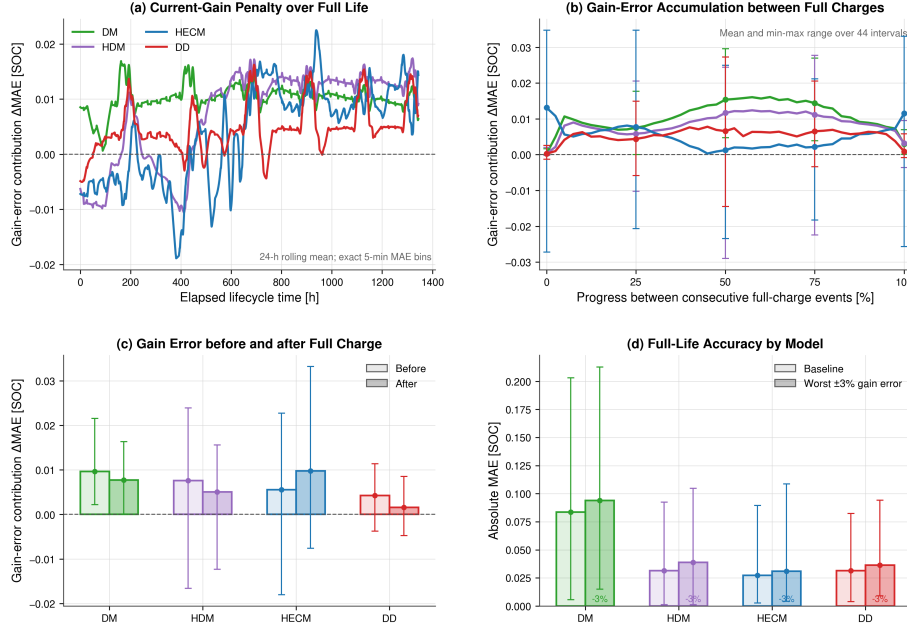


Figure 6: Current-gain-error diagnostic over the continuous C29 life replay. (a) Rolling 24-h gain-error contribution over elapsed life. (b) Mean and range over normalized intervals between consecutive full-charge events. (c) Penalty immediately before and after full-charge events. (d) Full-life baseline and adverse-gain MAE. Full-charge events reduce or restructure accumulated error but do not erase the lifetime penalty uniformly.

0.0002. A $+0.02\text{ V}$ voltage offset increases DD by 0.0038 and HDM by 0.0005, leaves DM effectively unchanged, and changes HECM by -0.0040 in these finite windows. The signed HECM result reflects compensation relative to its baseline trajectory. These offset results are included in the cross-scenario heatmap and in the family-balanced robustness synthesis.

4.4. Initialization error and recovery

Initialization performance uses the paired criterion defined above. The protocol maps the same -0.10 SOC-equivalent mismatch to the explicit DM/HDM Coulomb-counting state, the HECM EKF SOC state, and a Q_c offset of $-0.10C_{\text{nom}}$

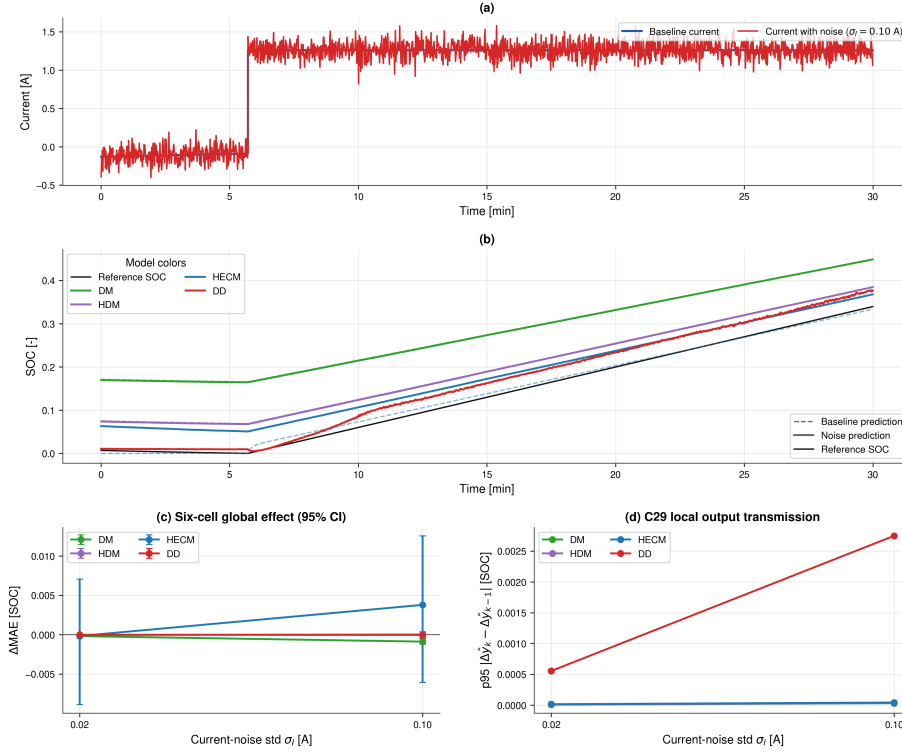


Figure 7: Noise sensitivity under additive current noise. (a) Baseline and noisy current in a reset-free 30 min C29 example. (b) Matched baseline and noisy SOC trajectories in the same window. (c) Cell-macro ΔMAE with hierarchical 95% confidence intervals across six holdout cells. (d) Local C29 transmission measured as the 95th percentile of the change in noise-induced output deviation between consecutive samples. The local panels illustrate the mechanism, while panel (c) provides the population-level comparison.

for DD. Figure 9 reports the resulting paired trajectory differences for these estimator-specific interventions. All four trajectories use source samples 2023 onward, while time remains referenced to the intervention at sample 0. The observed first-entry and persistent-recovery-or-censor means are 1.12 and 1.20 h for HECM and 1.29 and 2.60 h for DD. HECM and DD already satisfy the persistent endpoint at the first common observation in cell-macro fractions of 0.39 and 0.22, respectively. These endpoints are left-censored and enter the corresponding means as conservative upper bounds. DD’s 0.78 cell-macro relapse

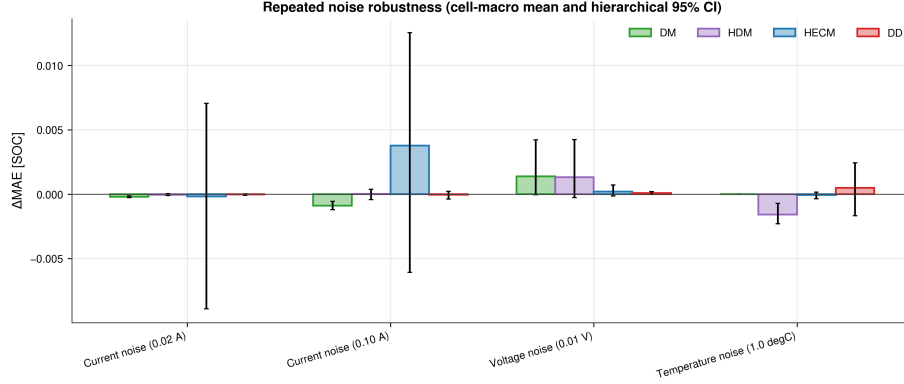


Figure 8: Repeated noise robustness across six holdout cells. Bars show cell-macro ΔMAE and error bars show hierarchical 95% confidence intervals with random seeds nested within cells. HECM is most affected by high current noise, while most other channel-model combinations remain close to baseline relative to the between-cell uncertainty.

fraction explains the difference between its first observed entry and persistent return, compared with 0.17 for HECM. DM first-entry and persistent recovery-or-censor times are both 22.28 h. HDM’s first-entry value of 17.21 h increases to 22.28 h under the persistent criterion because temporary entries do not constitute re-anchoring. The cell-macro persistent-censor fraction is 0.89 for both DM and HDM, compared with zero for HECM and DD. HECM also has the smallest excess-error area at 0.078 SOC h, compared with 0.116 SOC h for DD. Together, the observable paired trajectories, excess-error area, censoring, and relapse identify HECM as the strongest recovery implementation under this protocol. The common observation boundary defines the reported time resolution, while integration-based states generally require a later physical anchor.

4.5. Missing samples and signal integrity

Figure 10 compares the six-cell effects of periodic and random sample loss with those of timing jitter. Sample loss is substantially more consequential than timing jitter. Periodic freezing of every 50th sample increases MAE by 0.0072, 0.0065, 0.0028, and 0.0011 for DM, HDM, HECM, and DD, respectively. Random 2% loss produces corresponding increases of 0.0080, 0.0081, 0.0039, and

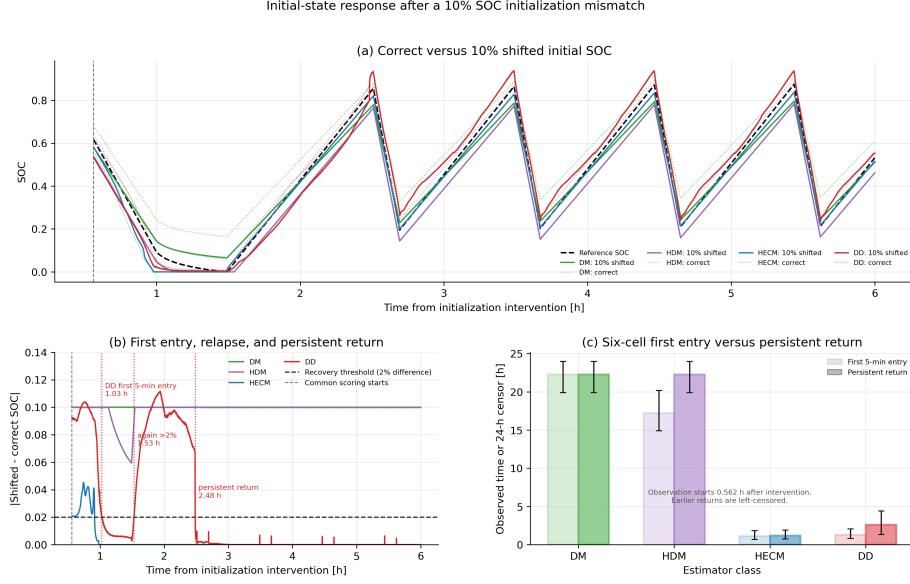


Figure 9: Response to a 10% SOC-equivalent initialization mismatch. (a) Correctly and incorrectly initialized trajectories for the dedicated C29 mid-life pair. (b) Absolute paired differences over the first 6 h. The DD trajectory enters the 2% band for 5 min, leaves it again, and only later returns persistently. The vertical gray marker denotes the first commonly scored sample at 0.562 h. (c) Observed first qualifying entry and persistent recovery-or-censor time across the six holdout cells with hierarchical 95% confidence intervals. Endpoints already satisfied at the common start are left-censored.

564 0.0026. By contrast, even the ± 0.9 s jitter case remains within approximately
565 6×10^{-4} SOC of baseline for every model, with confidence intervals generally
566 spanning zero. DD therefore has the smallest sample-loss penalty, while DM and
567 HDM retain their structural dependence on uninterrupted charge integration.

568 The one-hour burst dropout creates a distinct availability problem. Against
569 the duration-matched 48-h baseline, its macro Δ MAE is largest for HECM
570 (0.0239, 95% CI 0.0155–0.0366), followed by HDM (0.0209, 0.0098–0.0313) and
571 DM (0.0050, -0.0029 –0.0138). DD changes by -0.0093 (-0.0288 –0.0099), re-
572 flecting finite-window compensation between its baseline and frozen-input tra-
573 jectories. The interval is placed where the absolute unobserved net charge
574 is largest among starts that retain the required pre- and post-gap context.

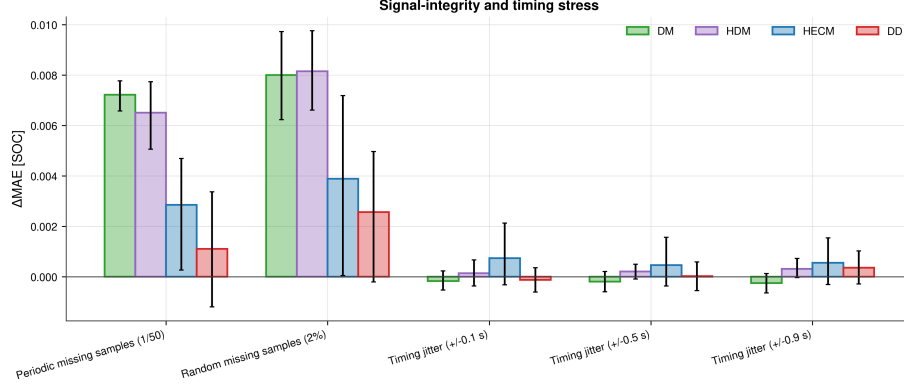


Figure 10: Six-cell signal-integrity and timing stress. Bars show cell-macro ΔMAE and hierarchical 95% confidence intervals for periodic and random sample loss and three timing-jitter levels. Sample loss clearly dominates timing jitter under the tested protocol.

Figure 11 compares each disturbed C29 output with its duration-matched no-dropout trajectory and reports the six-cell global effect. The paired local deviation has model-specific origins: DM/HDM retain the unobserved charge deficit until a physical anchor, HECM corrects through weak LFP voltage observability, and DD changes as its rolling input context is replaced by valid samples. Dropout contributes to the robustness dimension, while the recovery dimension uses the canonical paired initialization experiment.

4.6. Spike sensitivity

Single voltage spikes demonstrate why local and global metrics must be reported together. Full-window ΔMAE remains below 4×10^{-4} SOC for every estimator and is effectively zero for DM, HDM, and DD. Event-aligned analysis, however, identifies a reproducible DD transient: its mean spike-sample penalty across cells is approximately 7.7×10^{-3} SOC, compared with near-zero direct penalties for the other representatives. Figures 12 and 13 further show that the DD response varies across cells and SOH states and then decays after the event. This pattern is consistent with a one-sample voltage impulse entering both the absolute voltage and dU/dt channels and perturbing the rolling recurrent pre-

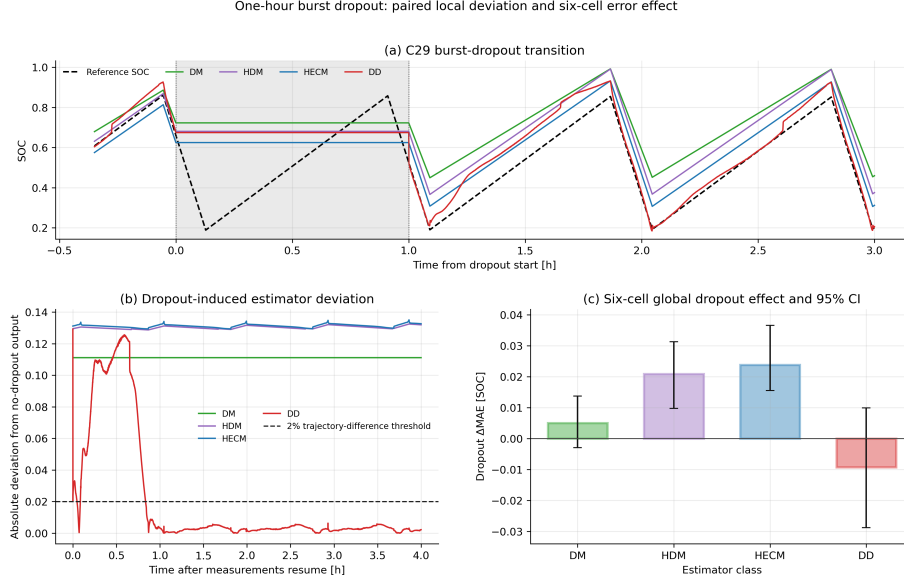


Figure 11: One-hour burst-dropout transition. (a) Estimator outputs around the shaded missing-input interval. (b) Absolute deviation from the paired duration-matched no-dropout output after measurements resume, with the 2% trajectory-difference line shown for context. (c) Six-cell global ΔMAE with hierarchical 95% confidence intervals.

diction. A channel-removal ablation would separate the individual contributions of the absolute and derivative inputs.

4.7. ADC quantization

Figure 14 compares representative raw and quantized current and voltage signals and summarizes the resulting six-cell ΔMAE . The tested ADC steps ($\Delta I = 0.01$ A, $\Delta U = 0.005$ V, and $\Delta T = 0.5$ °C) preserve the local signal shape but affect the tested representatives differently. DM and HDM increase by 0.0038 and 0.0035 MAE, respectively. HECM changes by -0.0018 and DD by $+0.0002$. All four confidence intervals overlap zero. At the selected coarse resolution, quantization produces smaller penalties than current gain error, sample loss, and burst dropout, while the integration-based representatives retain a measurable response.

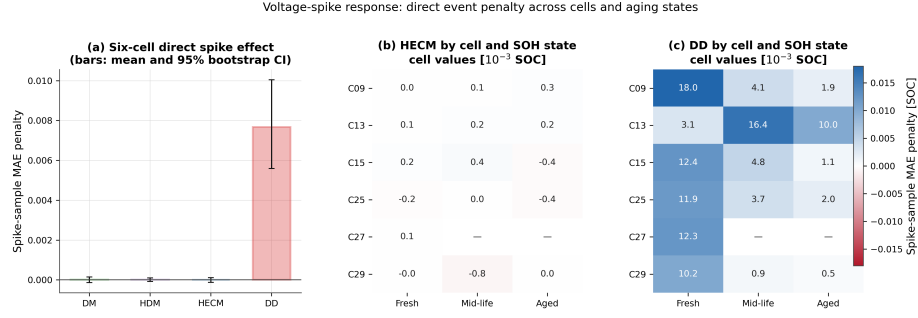


Figure 12: Voltage-spike sensitivity across six cells. (a) Mean event-sample MAE penalty with hierarchical 95% confidence intervals. (b,c) Cell- and SOH-state-resolved penalties for HECM and DD, respectively. Absent cell/state combinations are left blank. The event-aligned metric exposes the local DD response that is obscured in full-window Δ MAE.

4.8. Shared-SOH ablation

The primary comparison supplies one common causal LSTM-SOH trace to HDM, HECM, and DD. A paired ablation repeats selected scenarios with dataset reference SOH to isolate the influence of this shared service. Reference SOH reduces baseline MAE by 0.0370 for HDM and 0.0166 for HECM, while DD MAE increases by 0.0049 on average. The same direction persists for most noise, sample-loss, and dropout cases. At 3% current gain error, reference SOH improves HDM and HECM and changes DD by +0.0091. For temperature noise, the reference-minus-LSTM difference changes from its baseline value by 0.0016 for HDM and less than 10^{-4} for HECM and DD, indicating that the incremental temperature-noise response is largely independent of the shared SOH service. The reversed DD response reflects calibration and feature interaction with its SOH input. The primary LSTM-SOH results represent the complete causal estimator pipelines, while the reference-SOH runs explain how each SOC branch consumes health information.

4.9. Robustness synthesis across representatives

Figure 15 condenses 17 measurement-corruption and signal-integrity rows and shows how degraded-input behavior changes the nominal-accuracy pic-

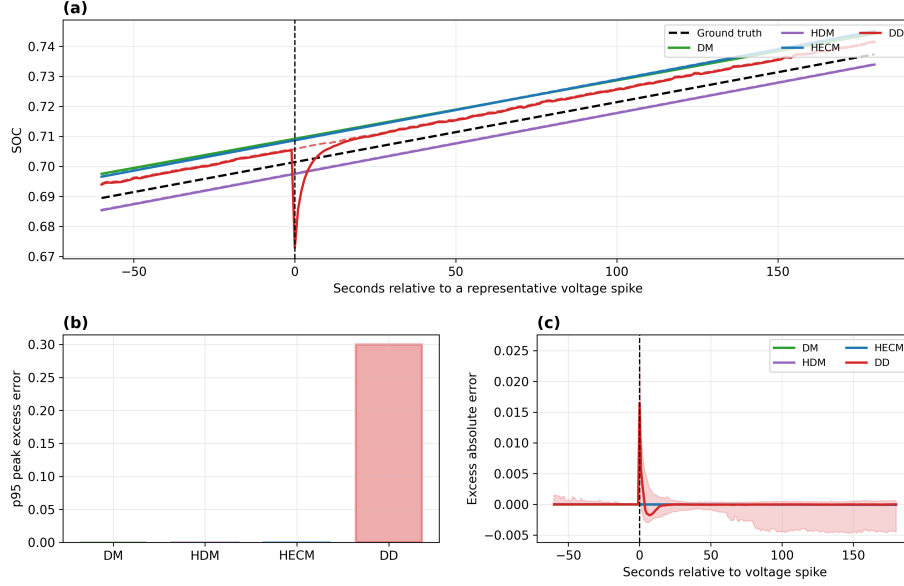


Figure 13: Event-aligned voltage-spike response. (a) Representative SOC trajectories around the injected spike, with the dashed black line showing dataset ground-truth SOC. (b) Model-wise 95th percentile peak excess error. (c) Mean excess absolute error aligned across repeated spikes, with the DD uncertainty envelope. The local response decays rapidly and is therefore weakly represented by full-window MAE.

ture. Persistent gain error and missing information create large global penalties. Current-gain sensitivity increases with magnitude for every representative in the matched signed sweep, periodic and random sample loss particularly affect DM and HDM, and the one-hour dropout produces the largest point penalty for HECM. Timing jitter remains nearly neutral. Voltage spikes reveal a distinct DD transient through event alignment despite their small full-window MAE. The alternative score aggregations then quantify how different priorities change the combined robustness order. Together, the heatmap and sensitivity analysis provide the scenario-specific evidence required for estimator selection.

Under the family-balanced definition, the robustness scores are 0.562 for DM, 0.468 for HDM, 0.428 for HECM, and 0.842 for DD. Equal weighting per evaluated scenario yields 0.574, 0.504, 0.560, and 0.781, while the highest-level-

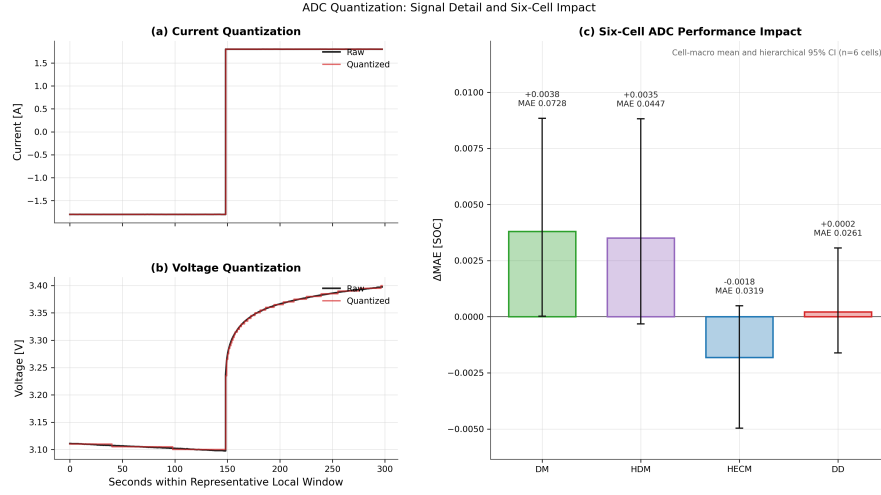


Figure 14: ADC quantization. (a,b) Representative raw and quantized current and voltage. (c) Six-cell Δ MAE with hierarchical 95% confidence intervals and resulting quantized-signal MAE annotations.

plus-offset variant yields 0.615, 0.512, 0.576, and 0.665. DD leads all three aggregations, while the changing score distances and HDM–HECM order demonstrate the influence of deployment priorities. The individual scenario results identify the disturbance mechanisms behind these aggregate changes.

4.10. Embedded performance benchmark

The STM32 experiment first verifies numerical equivalence before comparing speed or memory. Across the six nominal cell sequences, mean hardware–software MAE remains below 4×10^{-4} percentage points for every SOC core and reaches 3.41×10^{-6} percentage points for DD (Figure 17). These differences are orders of magnitude below the estimator-to-dataset errors and show that the float32 C implementations reproduce their software references sufficiently closely for deployment measurements.

The isolated DM and HDM SOC updates both have sub-microsecond median latency (0.171 and 0.165 μ s), HECM requires 23.9 μ s, and continuous-state DD requires 424.9 μ s (Figure 18). At the nominal 1-Hz update rate, these medians

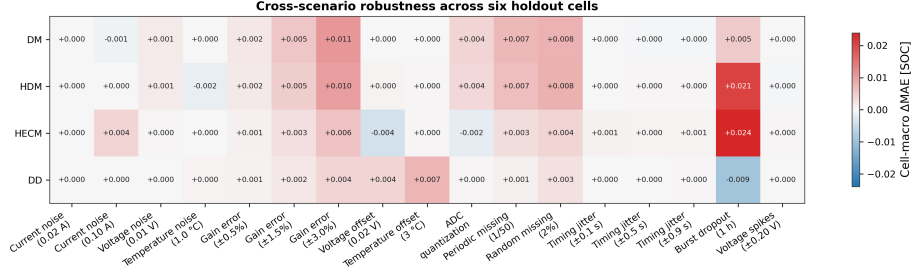


Figure 15: Cell-macro Δ MAE across 17 measurement-corruption and signal-integrity rows on six holdout cells. Current-gain rows use the adverse direction from the matched signed sweep. Positive values denote degradation relative to each model’s paired baseline, while small negative values arise from finite-window signed compensation. The initialization-mismatch response is evaluated separately in Figure 9.

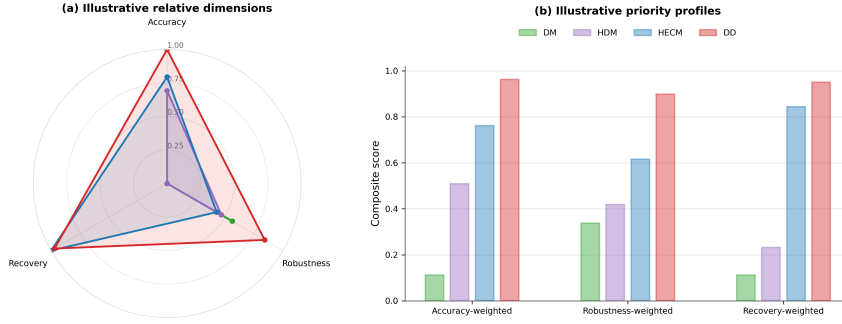


Figure 16: Decision-oriented synthesis for the four tested representatives. (a) Relative min-max-normalized accuracy, family-balanced robustness across eight evaluated disturbance families, and observed persistent paired recovery. The current-gain component uses the adverse direction from the matched signed sweep. (b) Composite scores under three 60/20/20 priority profiles. The profiles are interpreted together with the scenario-level evidence in Figure 15.

649 occupy less than 0.043% of one sampling period. This deadline share describes
650 the isolated estimator task and provides the execution margin available for inte-
651 gration with other BMS functions. The DD value in the cross-model comparison
652 is the continuous-state deployment variant. Exact 2024-sample rolling-window
653 semantics are evaluated separately because they recompute the full sequence at

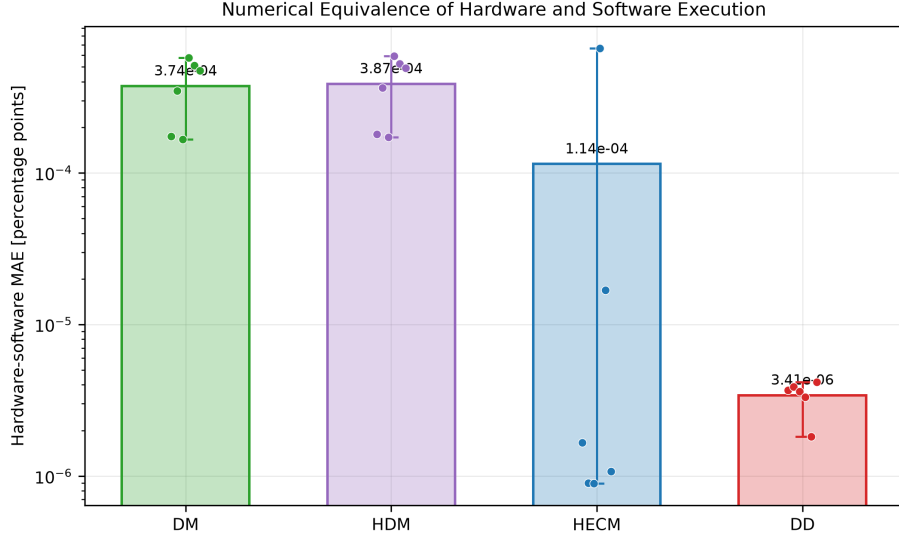


Figure 17: Numerical equivalence of STM32 and software execution over six cell sequences. Bars show mean hardware–software MAE in percentage points, dots show cells, and whiskers show the cell range. The logarithmic scale emphasizes that all implementation differences are negligible relative to model error.

every sample.

For the cross-model embedded performance comparison, DD denotes the continuous-state deployment variant used in the latency analysis. The compiled DM, HDM, HECM, and DD firmware occupies 27.0, 27.1, 470.1, and 115.9 KiB of flash, respectively. The corresponding measured peak runtime RAM from the representative C09 replay is 2.4, 2.4, 2.5, and 4.1 KiB (Figure 19). In bytes, DM and HDM each combine 872 B of statically allocated variable storage with a 320 B maximum dynamic-memory allocation and 1240 B of maximum call-stack use, yielding 2432 B. HECM reaches 2528 B because its static variable allocation increases to 968 B. Continuous-state DD combines 2588 B of static variable storage with 320 B of maximum dynamic-memory allocation and 1248 B of maximum call-stack use, yielding 4156 B. HECM flash occupancy is dominated by its parameter surfaces, whereas the DD parameter storage accounts for most of its flash demand. These values describe the SOC benchmark firmware with

On-Device Inference-Latency Distributions Across Six Cells

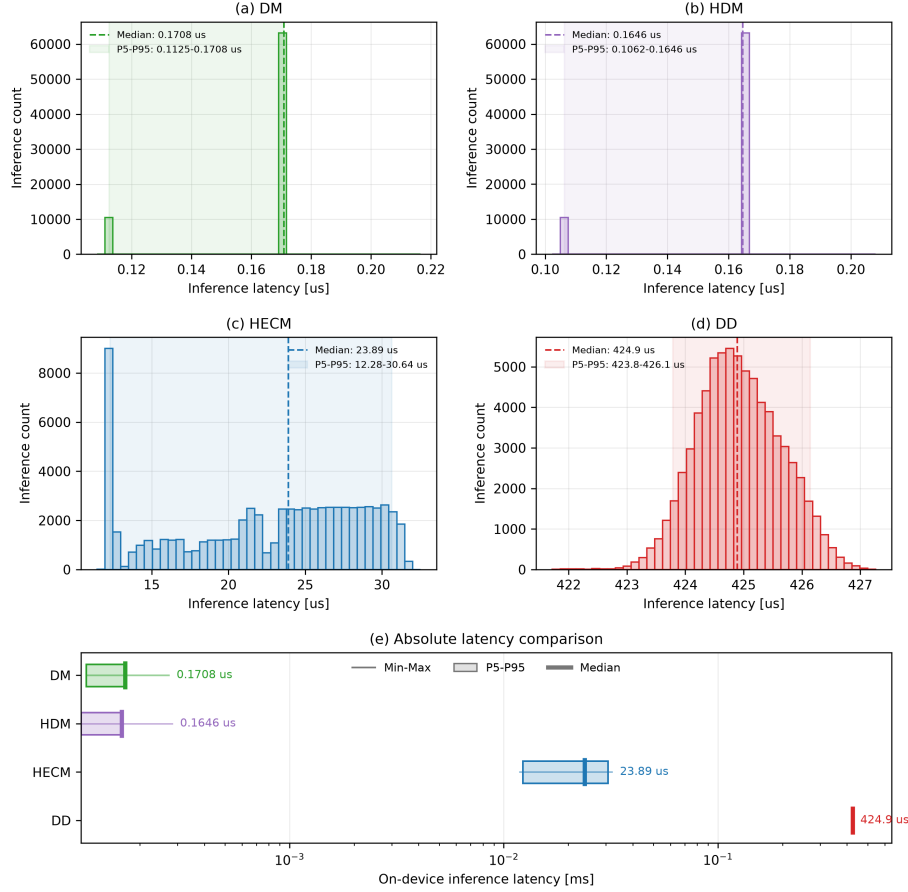


Figure 18: On-device SOC-core latency distributions across six cells and three replay rounds. Panels (a–d) show per-inference distributions. Panel (e) compares minimum, 5th percentile, median, 95th percentile, and maximum on a logarithmic scale. DD denotes continuous-state inference in this cross-model latency comparison.

668 the shared SOH trace provided externally and leave explicit resource margin for
 669 additional BMS tasks.

670 Figure 20 makes the DD deployment trade-off explicit, while Figure A.24
 671 provides the corresponding latency distributions over all six hardware cell se-
 672 quences. Exact rolling-window inference preserves the trained software seman-

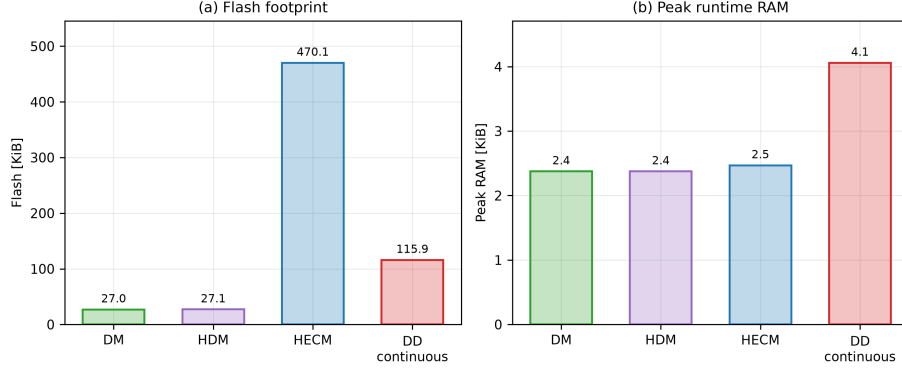


Figure 19: Embedded memory requirements of the isolated STM32 SOC implementations. (a) Compiled flash occupancy. (b) Peak runtime RAM measured during a representative 2025-sample C09 replay, combining statically allocated variable storage with maximum dynamic-memory allocation and call-stack use. DD denotes continuous-state execution, matching Figure 18. Exact rolling-window execution is compared separately in Figure 20.

tics but requires roughly 724 ms median inference time, or 72.4% of the one-second period, and 67.3 KiB peak runtime RAM. Continuous-state inference reduces latency to approximately 0.425 ms and peak RAM to 4.1 KiB while retaining similar error on the six nominal sequences. Periodically resetting that state has a comparable 4.1 KiB peak but creates much larger transient errors, with a maximum SOC error near 28 percentage points. Continuous-state execution therefore provides the practical hardware configuration. Its forwarded recurrent state defines a deployment mode that is evaluated separately from the primary rolling-window benchmark.

5. Discussion

5.1. Structural interpretation of the representatives

The four implementations expose distinct relationships between their update mechanisms and observed disturbance responses. DM has the largest nominal error and reacts strongly to persistent gain error and missing samples because its current-integration structure lacks voltage-residual correction. Its practical full-charge anchor limits accumulated drift, while the lifecycle experiment

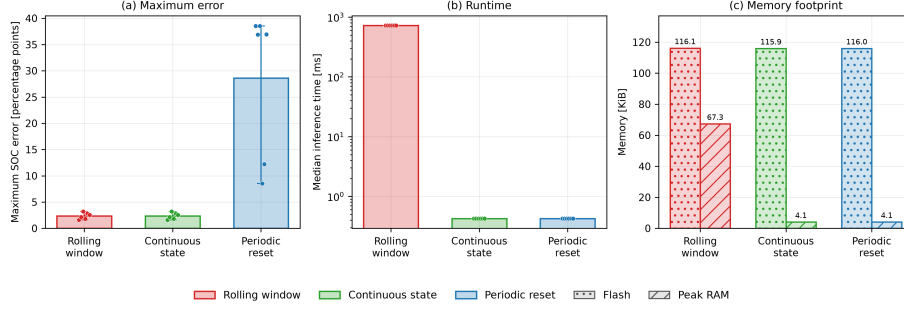


Figure 20: DD inference-mode trade-off on the STM32. The panels compare maximum dataset-SOC error, median latency, compiled flash occupancy, and measured peak runtime RAM for exact rolling-window, continuous-state, and periodically reset recurrent execution. Cell points show that periodic resets, not continuous forwarding, create the dominant accuracy loss.

shows that the reset restructures the gain-error contribution without removing it uniformly. HDM adapts the capacity denominator through learned SOH and thereby reduces capacity mismatch, yet it retains the same charge-integration core. Its similar response to current-gain error, missing charge information, and ADC quantization follows directly from this shared structure.

HECM combines current-driven state propagation with a voltage residual and has the strongest observed recovery profile under the tested initialization mapping. The common observation starts after 0.562 h and resolves subsequent re-anchoring behavior. High current noise and burst dropout produce its largest point penalties because corrupted or frozen inputs affect both state propagation and observer correction. The confidence interval for high current noise includes zero and shows substantial between-cell uncertainty. The dedicated lookup analysis shows that the lower HECM current-gain penalty is retained under systematic $\pm 10\%$ resistance scaling and ± 10 mV OCV shifts. The largest absolute macro interaction is 0.00026 MAE. Because the same perturbations affect baseline accuracy more visibly, lookup calibration remains relevant even though the incremental current-gain response is locally stable. This table dependence and the associated flash footprint form the central deployment trade-off

707 of HECM.

708 DD has the lowest nominal cell-macro error and the best adverse current-
709 gain score in this comparison. It is also comparatively resilient to isolated
710 sample loss. Event-aligned voltage spikes expose a recurrent transient that full-
711 window MAE hides. The selected voltage and derivative features and the rolling
712 recurrent context provide a plausible transmission path for this response and
713 identify a concrete target for feature ablation and disturbance-aware training.

714 5.2. *Why nominal accuracy is insufficient*

715 Nominal MAE describes average agreement with the dataset ground truth
716 under clean inputs. The disturbance and recovery experiments add the DD
717 voltage-spike transient, HECM re-anchoring behavior, shared-SOH feature cou-
718 pling, and model-specific responses to gain error and missing information. Em-
719 bedded measurements add latency, memory, and inference-state semantics. These
720 dimensions create a deployment profile for each implementation, while the weight-
721 ing analysis shows how application priorities reshape their combined scores.
722 Estimator selection can therefore begin with the relevant fault and recovery
723 requirements and then incorporate the available compute, memory, and state-
724 service architecture.

725 5.3. *Implications for BMS deployment*

726 Among the tested representatives, DD has the lowest nominal error against
727 the dataset ground truth and remains a strong candidate when its recurrent
728 implementation cost and voltage-spike response fit the application. HECM is
729 preferable when persistent correction after a state mismatch has high priority
730 and a larger flash image is available. DM and HDM remain attractive when
731 deterministic sub-microsecond execution and small memory dominate. Their
732 integration structure places correspondingly greater requirements on current
733 calibration, reliable sampling, and full-charge anchoring. These mechanism-
734 based trade-offs connect the benchmark results directly to BMS deployment
735 priorities.

736 The STM32 results connect algorithmic behavior with deployment cost. All
737 four isolated SOC cores meet the one-second deadline in the tested replay, while
738 their timing margins span more than six orders of magnitude between DM/HDM
739 and exact rolling-window DD. Continuous-state DD removes most of the rolling-
740 window cost by forwarding its recurrent state. The firmware measurements
741 include flash, maximum call-stack use, dynamic-memory demand, and numerical
742 equivalence. Integration of the shared SOH-LSTM, energy measurement, pack
743 communication, and concurrent BMS scheduling forms the next system-level
744 validation stage. Table 3 relates the measured estimator profiles to concrete
745 deployment priorities.

746 Figure 16 complements the recommendation map by combining the mea-
747 sured accuracy, robustness, and recovery dimensions into relative priority pro-
748 files. The three profiles assign 60% to their named dimension and 20% to each
749 remaining dimension. Family-balanced, equal-scenario, and high-severity ag-
750 gregations produce different score distances and change the HDM-HECM or-
751 der. This sensitivity turns the score into a configurable decision aid, while the
752 scenario heatmap identifies the disturbance-specific strengths and weaknesses
753 behind each profile.

754 5.4. Limitations and future extensions

755 The conclusions are grounded in four specified implementations and six hold-
756 out cells from one controlled LFP aging dataset. Alternative reset logic, ECM
757 structures, neural architectures, features, and training data can produce differ-
758 ent estimator profiles. The low- and middle-load strata contain multiple cells,
759 whereas C29 supplies one exploratory high-load trajectory and C27 contributes
760 no mid-life or aged window. The reported confidence intervals therefore quan-
761 tify the available between-cell variation within this design. Transfer to other
762 chemistries, cell formats, pack configurations, thermal environments, and field
763 duty cycles requires corresponding cell-disjoint validation. Nominal accuracy is
764 referenced to one consistently reconstructed offline SOC ground truth based on
765 Coulomb counting and voltage anchors.

Table 3: Decision-oriented recommendation map derived from the benchmark results. The table identifies the strongest observed representative under one dominant deployment priority and the stated protocol boundaries.

Primary deployment priority	Selected representative	Main supporting evidence	benchmark	evi-	Main trade-off to consider
Lowest nominal SOC error	DD	Lowest six-cell macro (0.0258) and RMSE (0.0300)	MAE		Rolling-window execution is the most computationally expensive implementation
Strongest observed recovery from initialization mismatch	HECM	Lowest observed persistent-recovery-or-censor time and substantially fewer first-entry relapses than DD	persistent-	time and	Recovery observation begins at the common 0.562-h boundary after estimator-specific interventions
Lowest adverse current-gain penalty	DD	Matched signed Δ MAE 0.0038 at 3%, below HECM, HDM, and DM	sensitivity	gives	Temperature-offset and local voltage-spike sensitivity remain visible
Smallest observed periodic/random sample-loss penalty	DD	Δ MAE 0.0011/0.0026			The duration-matched dropout effect is non-positive but highly variable across cells
Lowest deterministic compute cost	DM/HDM	Sub-microsecond median SOC-core latency and approximately 27 KiB flash	median SOC-core		Integration structure is sensitive to gain error and missing charge information
Practical fast neural deployment	DD continuous state	Approximately 0.425 ms median and 4.1 KiB peak runtime RAM	median		Different recurrent semantics from the primary rolling-window benchmark

766 The primary cross-model campaign concentrates on the BMS measurement
 767 chain, signal availability and timing, and deployment-accessible initialization
 768 states. The separate HECM lookup analysis probes uniform local resistance and
 769 OCV deviations, while broader parameter mismatch, pack-level interactions,
 770 and supervisory BMS faults form a separate validation layer. The initialization
 771 experiment maps an SOC-equivalent offset into each estimator’s available state
 772 and observes recovery from the common 0.562-h boundary. These times char-
 773 acterize operational re-anchoring under class-specific interventions rather than
 774 identical latent-state convergence. The continuous C29 replay complements the
 775 six-cell windows with a mechanism-focused lifecycle view. The hourly SOH ser-
 776 vice represents gradual capacity fade, and channel-removal experiments would
 777 further isolate the DD voltage-spike transmission path. HECM uses HPPC-
 778 derived parameter surfaces identified from the development cells and held fixed
 779 throughout evaluation. The sensitivity analysis therefore characterizes local ro-
 780 bustness around this calibrated model rather than arbitrary lookup-table error.

781 The hardware profile covers the isolated SOC firmware with SOH supplied
782 as an external causal trace. Runtime-RAM watermarks come from the repre-
783 sentative C09 replay, and the rolling DD stack maximum is based on its valid
784 post-warm-up calls. The next hardware stage combines the SOC and SOH
785 services, measures energy demand, and evaluates estimator scheduling along-
786 side communication and supervisory BMS tasks. Broader data campaigns can
787 then extend the disturbance protocol to interacting faults, additional operating
788 environments, and pack-level closed-loop behavior.

789 6. Conclusion

790 This study establishes a common robustness and embedded-performance
791 benchmark for four structurally different SOC-estimation implementations. DD
792 achieves the lowest nominal cell-macro error and the strongest results across sev-
793 eral disturbance families, while local and hardware-specific analyses reveal its
794 voltage-spike transient and rolling-window cost. HECM provides the strongest
795 observed recovery profile and voltage-residual correction, together with the
796 largest flash demand and pronounced responses to current noise and prolonged
797 dropout. Its lower current-gain penalty remains stable under the tested local
798 resistance and OCV lookup perturbations, although lookup calibration con-
799 tinues to affect nominal accuracy. DM and HDM provide deterministic sub-
800 microsecond execution with minimal memory, while their current-integration
801 structure increases sensitivity to gain error, missing charge information, and
802 physical re-anchoring.

803 The combined evidence turns estimator selection into a deployment-oriented
804 comparison of accuracy, disturbance-specific robustness, recovery, latency, flash,
805 and runtime RAM. The STM32 measurements confirm numerical equivalence
806 with the software implementations and one-second execution of all isolated SOC
807 cores, while the DD deployment variants show how recurrent-state handling
808 changes both cost and transient behavior. Scenario-level evidence remains cen-
809 tral because the preferred implementation changes with the sensing fault profile,

810 recovery requirement, and available hardware margin. This framework provides
811 a direct basis for extending validation to complete SOC-SOH execution, concur-
812 rent BMS tasks, additional operating environments, and pack-level applications.

813 Appendix A. Dataset Split, Holdout Coverage, and Evaluation Pro- 814 tocol

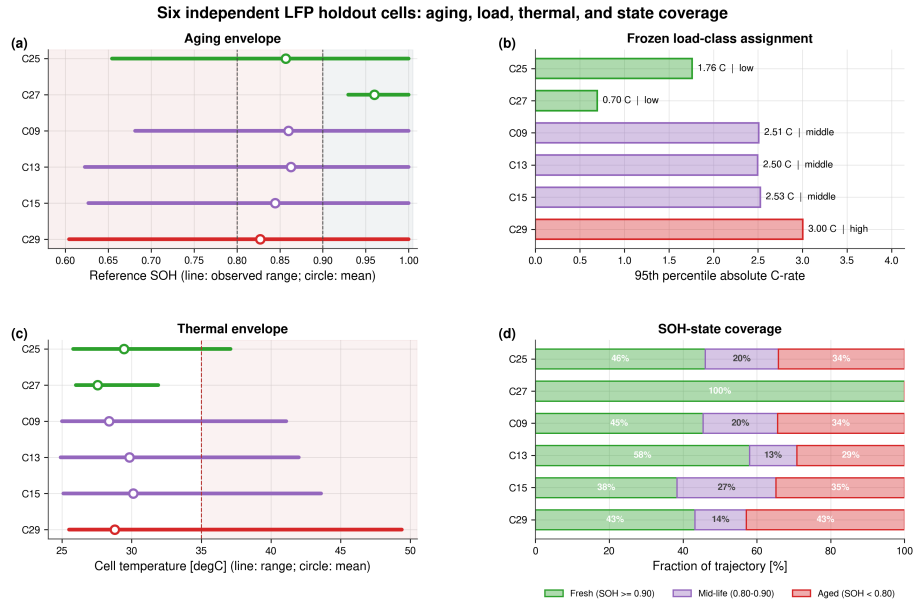


Figure A.21: Quantitative coverage of the six independent holdout cells. Panels show observed SOH, the descriptive load-class assignment, thermal envelope, and full-trajectory SOH-state coverage. High-load evidence consists only of C29 and is exploratory.

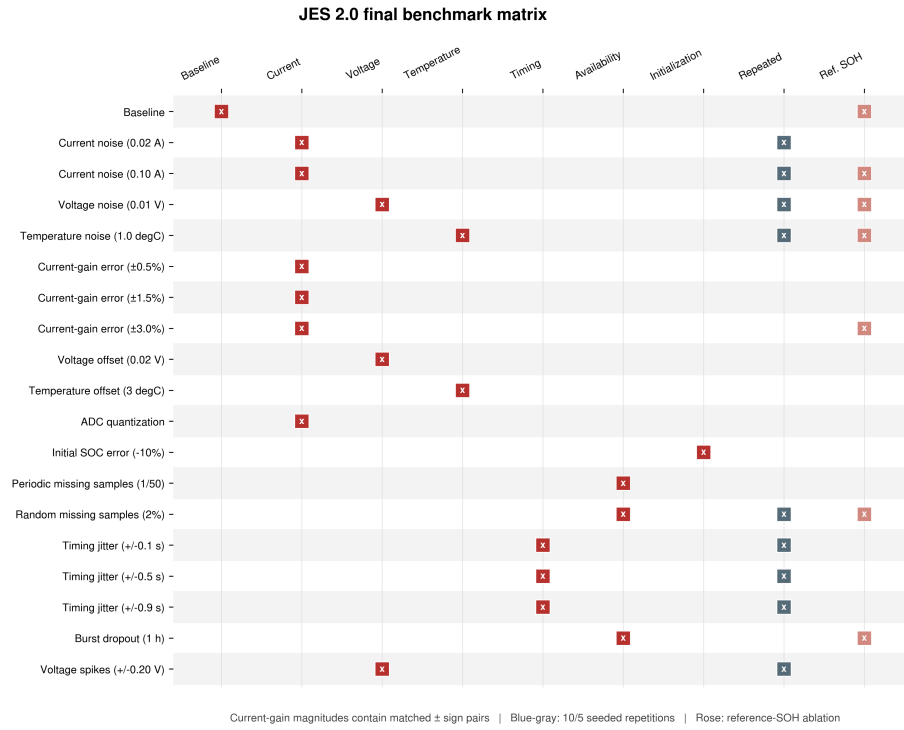


Figure A.22: Complete 19-scenario benchmark matrix. Red markers identify the manipulated measurement, timing, availability, or initialization channel. Blue-gray markers denote repeated stochastic cases. Rose markers identify paired reference-SOH ablations. Each current-gain magnitude contains matched positive and negative sign sublevels.

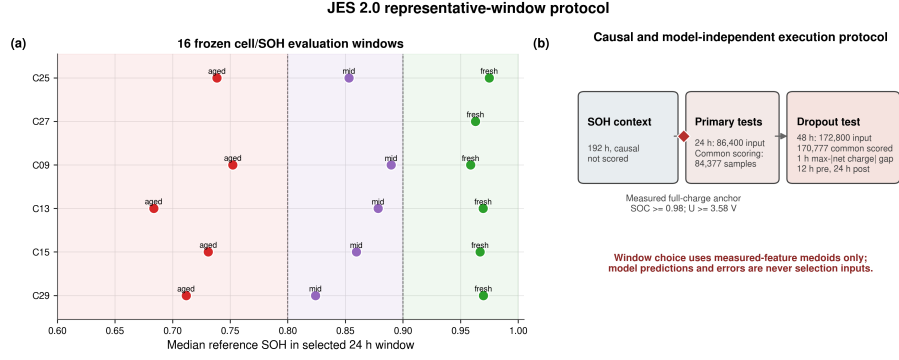


Figure A.23: Representative-window protocol. Sixteen cell/SOH windows are selected using measured-feature medoids. Every estimator is scored from source sample 2023, yielding 84,377 matched points per 24-h window after a 192-h causal SOH context. The dropout case uses 48 h, places the gap at the eligible one-hour interval with maximum absolute net charge, and contributes 170,777 matched points.

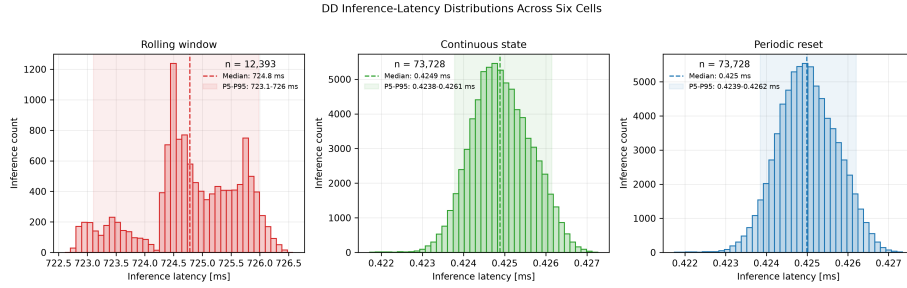


Figure A.24: DD latency distributions over the six hardware cell sequences. Exact rolling-window execution is shown separately from continuous-state and periodically reset execution because their scales and recurrent semantics differ substantially.

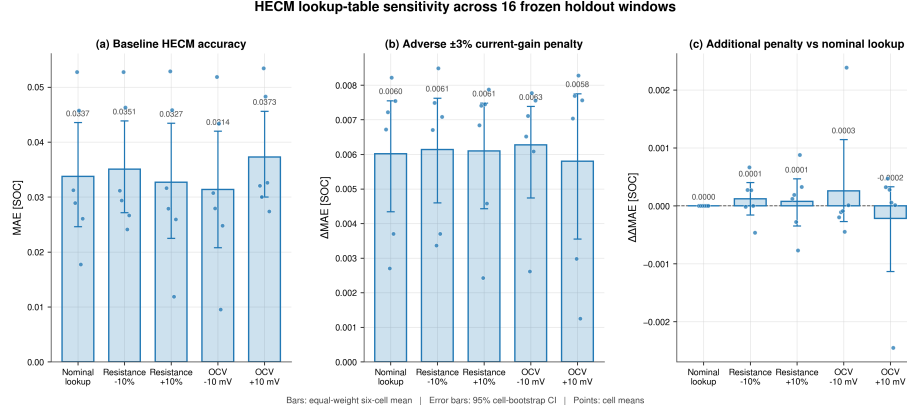


Figure A.25: HECM lookup-table sensitivity over all 16 frozen windows. (a) Baseline MAE under the nominal lookup, joint $\pm 10\%$ scaling of R_i , R_1 , and R_2 , and uniform ± 10 mV OCV shifts. (b) Adverse current-gain penalty from matched -3% and $+3\%$ gain errors relative to the baseline under the same lookup condition. (c) Lookup-gain interaction relative to the nominal-table penalty. Bars show equal-weight six-cell means, error bars show 95% cell-bootstrap intervals, and points show cell means.

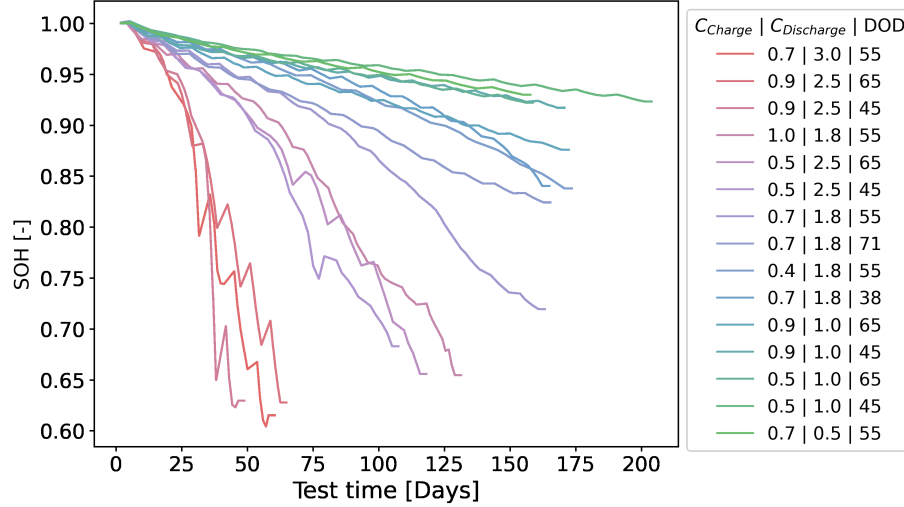


Figure A.26: Measured SOH trajectories across the complete LFP aging campaign. The legend reports the applied charge C-rate, discharge C-rate, and depth of discharge for each trajectory. The plotted SOH values originate from repeated capacity checkups and are interpolated between checkup anchors. The fixed training, validation, and holdout assignments and their measured coverage are reported in Table A.4.

Table A.4: Measured coverage of all cells in the fixed model-development and holdout split. The 95th-percentile absolute C-rate is calculated over each complete 1-Hz trajectory. Load classes are defined only for the six holdout cells. The high-load class contains C29 only and is exploratory.

Cell	Split	Duration [d]	SOH range	T range [$^{\circ}\text{C}$]	P95 $ C_{\text{rate}} $	Holdout load
C01	Training	198.6	0.922–1.000	24.3–29.4	1.01	–
C03	Training	153.3	0.921–1.000	25.9–30.7	1.00	–
C05	Training	166.1	0.917–1.000	25.2–30.6	1.00	–
C11	Training	113.2	0.656–1.000	25.6–42.8	2.48	–
C17	Training	158.7	0.719–1.000	25.8–34.9	1.75	–
C23	Training	167.8	0.837–1.000	26.1–37.2	1.74	–
C07	Validation	167.6	0.874–1.000	25.2–32.0	1.01	–
C19	Validation	160.4	0.840–1.000	26.3–33.9	1.75	–
C21	Validation	160.8	0.824–1.000	26.2–37.6	1.75	–
C09	Holdout	102.6	0.682–1.000	25.0–41.1	2.51	Middle
C13	Holdout	44.7	0.623–1.000	24.9–42.0	2.50	Middle
C15	Holdout	60.7	0.627–1.000	25.1–43.6	2.53	Middle
C25	Holdout	127.1	0.655–1.000	25.8–37.1	1.76	Low
C27	Holdout	152.0	0.930–1.000	26.0–31.9	0.70	Low
C29	Holdout	56.0	0.604–1.000	25.5–49.4	3.00	High*

Table B.5: Six-cell baseline cell-macro statistics with hierarchical 95% confidence intervals.

Estimator	MAE [SOC]	RMSE [SOC]	P95 error [SOC]
DM	0.0690 [0.0499, 0.0823]	0.0740 [0.0532, 0.0881]	0.1088 [0.0768, 0.1307]
HDM	0.0412 [0.0301, 0.0515]	0.0442 [0.0319, 0.0557]	0.0649 [0.0449, 0.0846]
HECM	0.0337 [0.0246, 0.0434]	0.0388 [0.0286, 0.0493]	0.0644 [0.0468, 0.0821]
DD	0.0258 [0.0189, 0.0325]	0.0300 [0.0223, 0.0366]	0.0508 [0.0375, 0.0619]

Table B.6: Exact paired baseline-MAE comparisons across the six holdout cells. Differences are defined as estimator B minus estimator A. Holm correction is applied across the six model-pair tests.

Estimator A	Estimator B	Mean difference [SOC]	95% CI	d_z	$p_{\text{exact}} / p_{\text{Holm}}$
DM	HDM	-0.0278	[-0.0381, -0.0171]	-1.93	0.0313 / 0.1875
DM	HECM	-0.0352	[-0.0470, -0.0224]	-2.00	0.0313 / 0.1875
DM	DD	-0.0431	[-0.0558, -0.0286]	-2.29	0.0313 / 0.1875
HDM	HECM	-0.0075	[-0.0162, 0.0015]	-0.61	0.1563 / 0.1875
HDM	DD	-0.0154	[-0.0243, -0.0084]	-1.39	0.0313 / 0.1875
HECM	DD	-0.0079	[-0.0176, -0.0005]	-0.68	0.0938 / 0.1875

Table B.7: Selected cell-macro scenario penalties relative to paired baseline. Current-gain values use the adverse sign from the matched $\pm 3\%$ sweep.

Scenario	DM	HDM	HECM	DD
Current noise, $\sigma_I = 0.10$ A	-0.0009	+0.0000	+0.0038	-0.0000
Current gain, adverse 3%	+0.0108	+0.0101	+0.0060	+0.0038
Voltage offset, +0.02 V	+0.0000	+0.0005	-0.0040	+0.0038
Temperature offset, +3 °C	+0.0000	+0.0002	+0.0002	+0.0074
ADC quantization	+0.0038	+0.0035	-0.0018	+0.0002
Random missing samples, 2%	+0.0080	+0.0081	+0.0039	+0.0026
Timing jitter, ± 0.9 s	-0.0003	+0.0003	+0.0006	+0.0003
Burst dropout, 1 h	+0.0050	+0.0209	+0.0239	-0.0093
Voltage spikes, ± 0.20 V	+0.0000	-0.0003	+0.0002	-0.0001

Table B.8: Sensitivity of the illustrative normalized robustness score to the aggregation rule.

Estimator	Family balanced	Equal scenario	Highest level plus offsets
DM	0.562	0.574	0.615
HDM	0.468	0.504	0.512
HECM	0.428	0.560	0.576
DD	0.842	0.781	0.665

Table B.9: Canonical paired initialization recovery and target-hardware summary. Recovery values are equal-weight cell means on the common source-sample interval, which begins 0.562 h after the intervention. Endpoints already satisfied at this boundary are recorded at the boundary as conservative upper bounds.

Metric	DM	HDM	HECM	DD
First 300-s entry/censor time [h]	22.28	17.21	1.12	1.29
Persistent recovery/censor time [h]	22.28	22.28	1.20	2.60
Recovery excess-error AUC [SOC h]	1.735	1.738	0.078	0.116
First-entry censored fraction	0.89	0.67	0.00	0.00
Persistent-recovery censored fraction	0.89	0.89	0.00	0.00
Relapse fraction after first hold	0.00	0.22	0.17	0.78
Median SOC-core latency [μ s]	0.171	0.165	23.9	424.9 ^a
Flash [KiB]	27.0	27.1	470.1	115.9 ^a
Peak runtime RAM [KiB]	2.4	2.4	2.5	4.1 ^a

^aContinuous-state DD. Exact rolling-window execution requires approximately 724 ms and 67.3 KiB peak RAM.

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821 The author contribution statement is provided in the non-blinded submission
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823 **Declaration of competing interest**

824 The authors declare that they have no known competing financial interests or
825 personal relationships that could have appeared to influence the work reported
826 in this paper.

827 **Declaration of generative AI and AI-assisted technologies** 828 **in the manuscript preparation process**

829 During the preparation of this work the author(s) used ChatGPT, DeepL,
830 and Grammarly in order to improve the linguistic quality and style of the
831 manuscript. After using these tools, the author(s) reviewed and edited the
832 content as needed and take(s) full responsibility for the content of the published
833 article.

834 **Data availability**

835 The dataset is publicly available through its cited TU Berlin repository
836 record [28]. Code, model configurations, benchmark definitions, random seeds,
837 result summaries, hardware timing records, and analysis procedures are main-
838 tained in a public GitHub repository. The repository contains the neural model
839 assets and HECM parameterization used for the reported experiments.

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